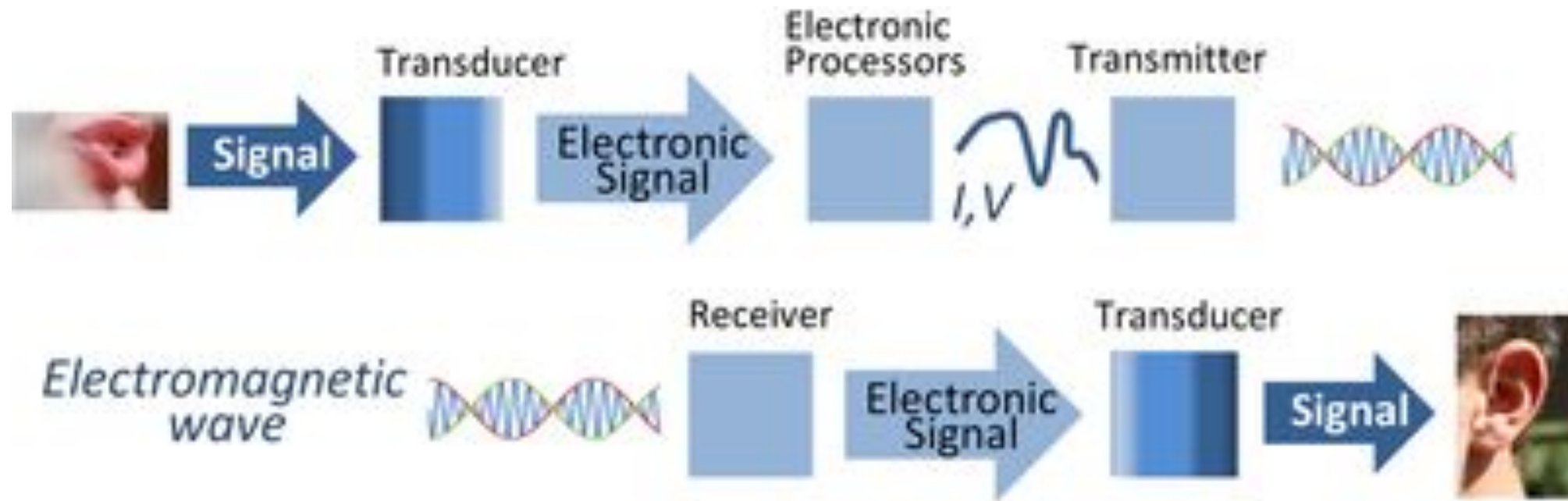


Fundamentals of Computer Networks and Management

Atul Kahate

Sessions 1 and 2

Electronic Communication System



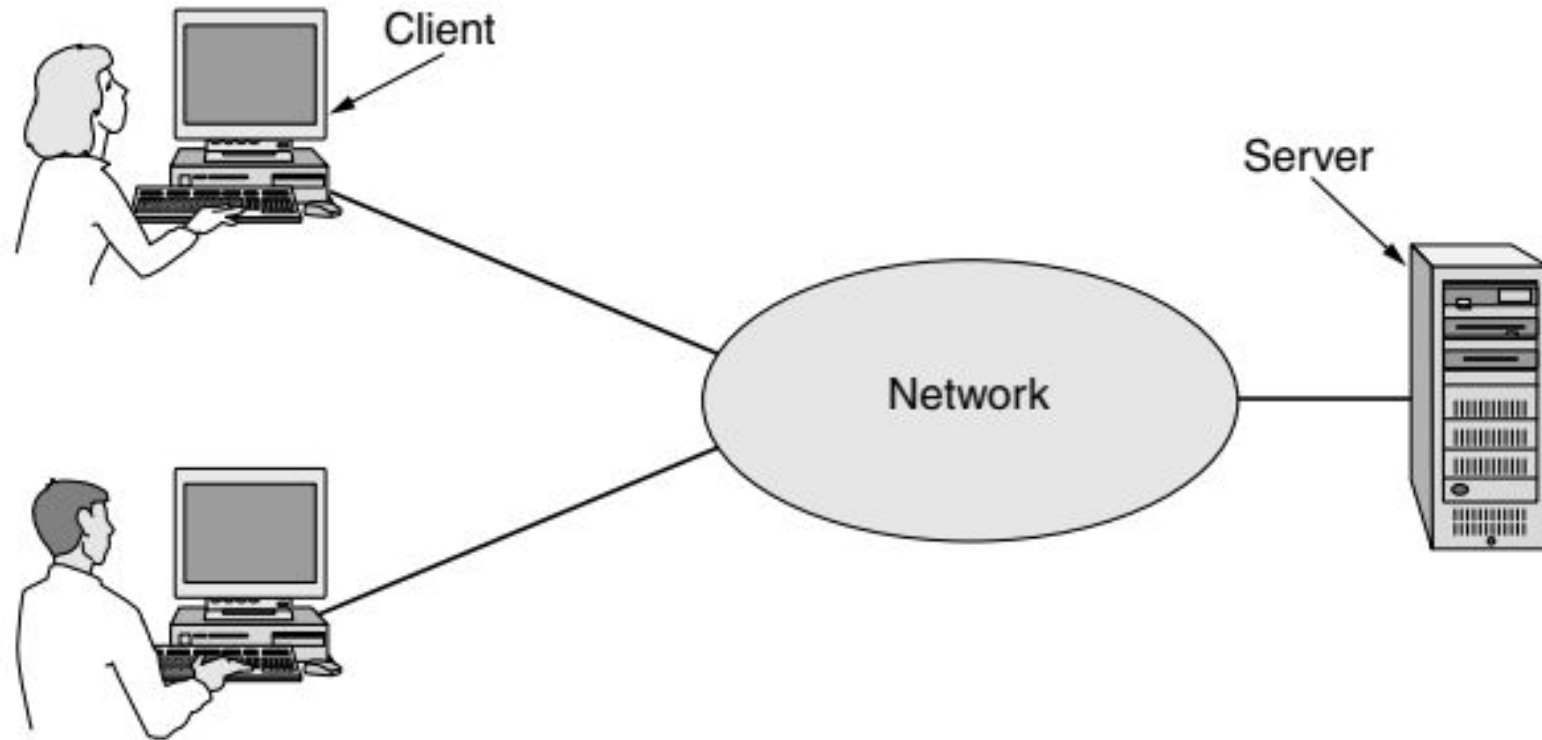
Introduction to Communication System

- **Communication system** allows exchange of data between two stations, **transmitter** and **receiver**
- **Signals** pass from **source** to **destination** through what is called **channel**, which represents a way that signal use it to move from source toward destination
- To transmit signals in communication system, it must be first processed in several stages, beginning from **signal representation**, to **signal shaping** until **encoding** and **modulation**
- After preparing the transmitted signal, it is passed to the **transmission line** of channel
- Due to the signal crossing this media it faces many impairments such as **noise**, **attenuation** and **distortion**

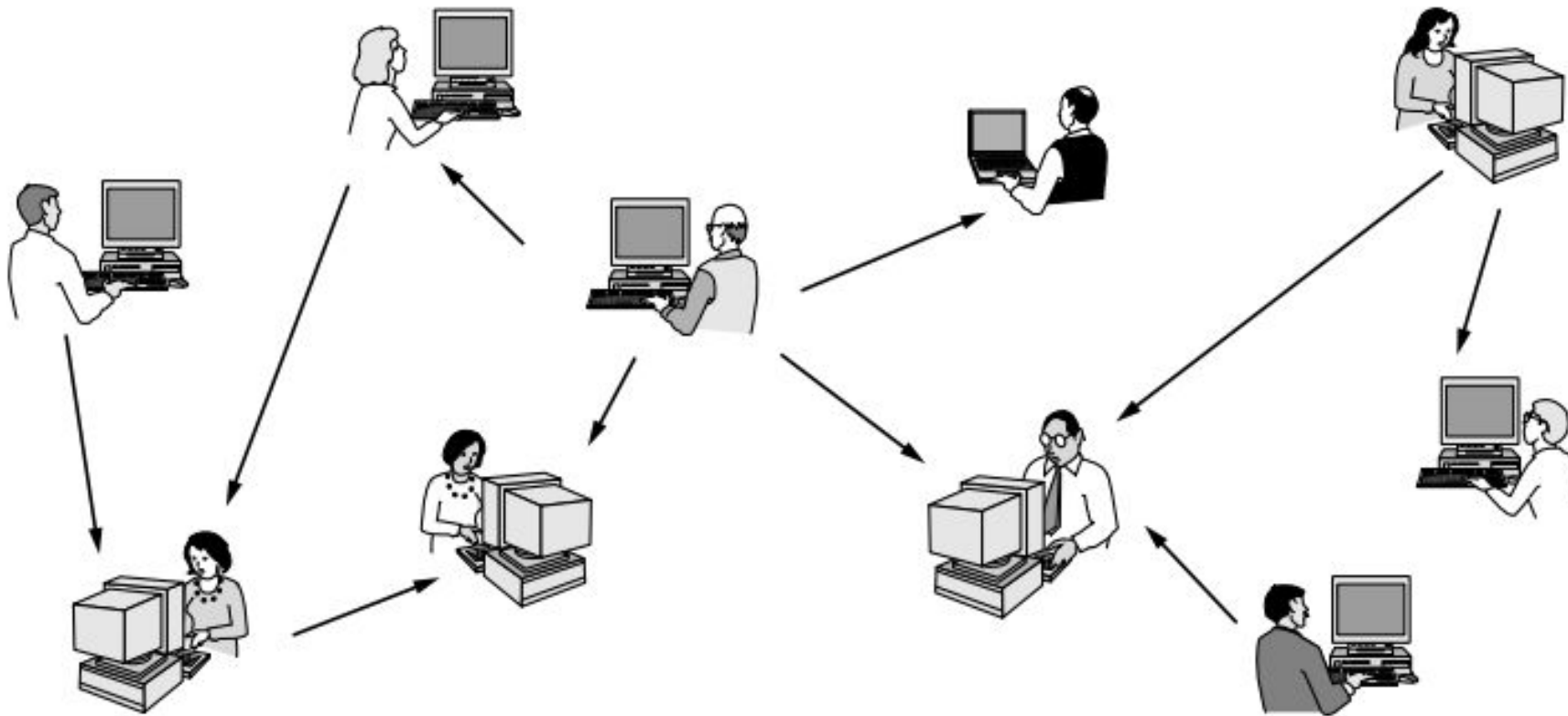
Components of a Data Communication System

- Message
 - The data/information to be transmitted. It can be in the form of text, audio, video, photos, or combination of them.
- Transmission device or sender
 - A device that actually sends out the message.(eg computer , telephone, etc).
- A medium
 - A physical channel where the information travels from the sender to the receiver.
- Receiving device or receiver
 - A device that receives the message then process the information in the message.

Client-Server Model



Peer-to-Peer (P2P) System

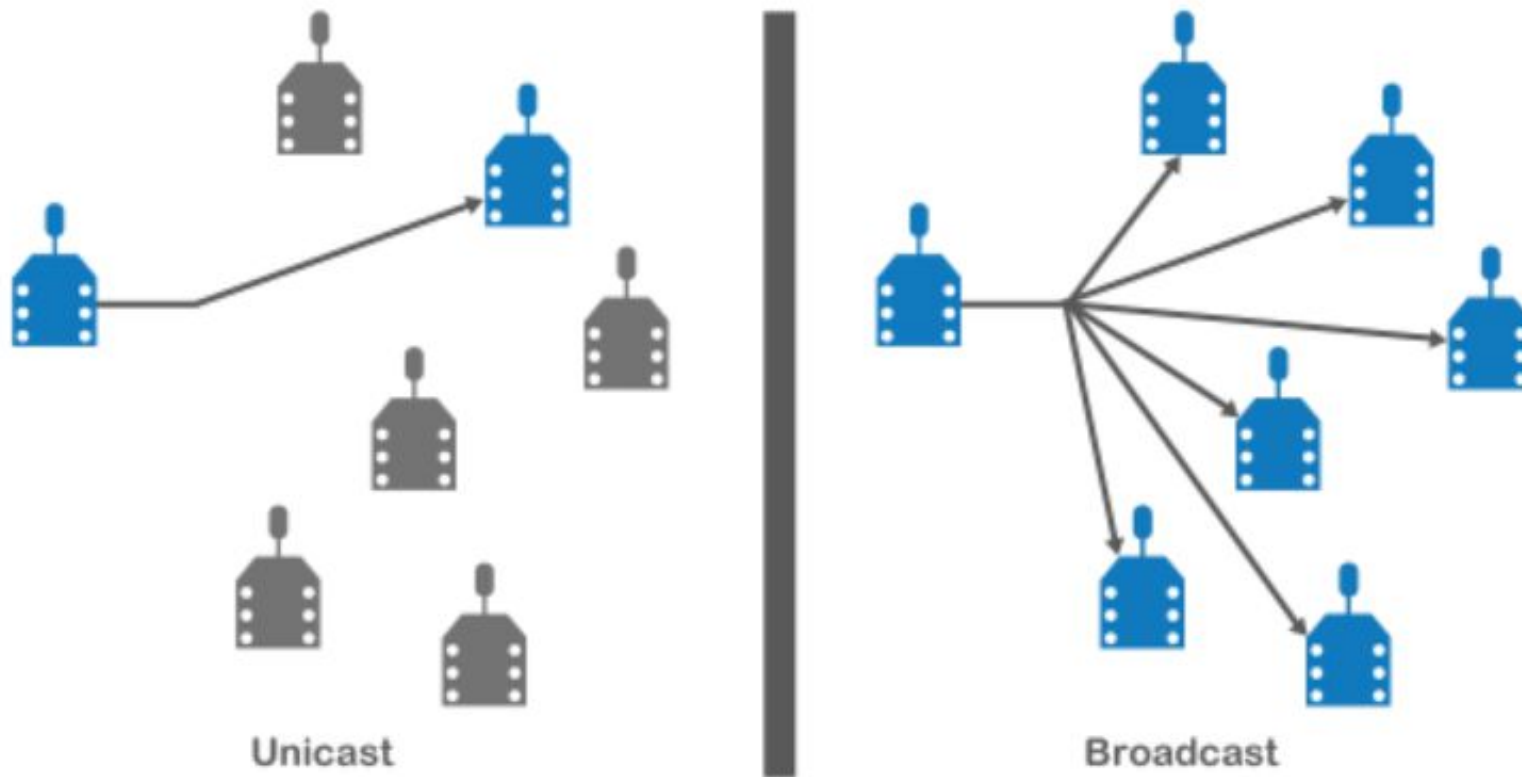


Sessions 3 and 4

Network Hardware and Software

- There are two types of transmission technology: **broadcast links** and **point-to-point links**. (See next slide)
- Point-to-point links connect individual pairs of machines. To go from the source to the destination on a network made up of point-to-point links, short messages, called packets, may have to first visit one or more intermediate machines. Often multiple routes, of different lengths, are possible, so finding good ones is important in point-to-point networks. Point-to-point transmission with exactly one sender and exactly one receiver is sometimes called **unicasting**.
- In contrast, on a broadcast network, the communication channel is shared by all the machines on the network; packets sent by any machine are received by all the others. An address field within each packet specifies the intended recipient. Upon receiving a packet, a machine checks the address field. If the packet is intended for the receiving machine, that machine processes the packet; if the packet is intended for some other machine, it is just ignored. A **wireless network** is a common example of a broadcast

Unicast versus Broadcast



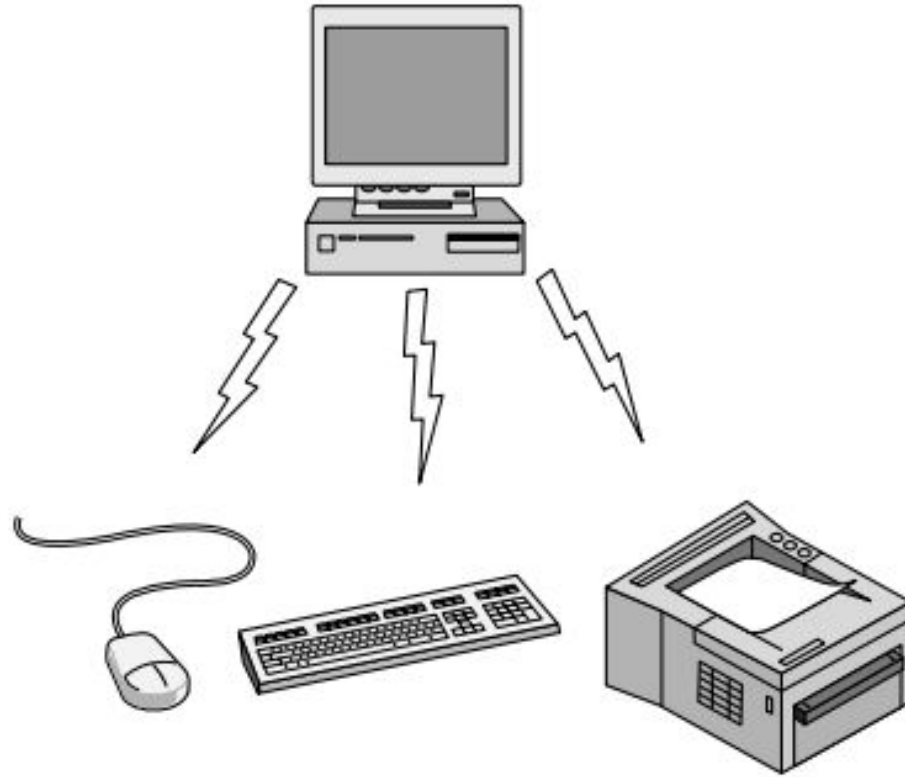
Scale and Network Types

Interprocessor distance	Processors located in same	Example
1 m	Square meter	Personal area network
10 m	Room	Local area network
100 m	Building	
1 km	Campus	
10 km	City	Metropolitan area network
100 km	Country	Wide area network
1000 km	Continent	
10,000 km	Planet	The Internet

Personal Area Network (PAN)

- A **Personal Area Network (PAN)** lets devices communicate over the range of a person
- A common example is a wireless network that connects a computer with its peripherals
- Some companies got together to design a short-range wireless network called **Bluetooth** to connect these components without wires
- The idea is that if your devices have Bluetooth, then you need no cables
- In the simplest form, Bluetooth networks use the **master-slave** concept
- The system unit is the master and keyboard, etc., are slaves
- The master tells the slaves what addresses to use, when they can broadcast, etc. (See figure)

Bluetooth



Local Area Network (LAN)

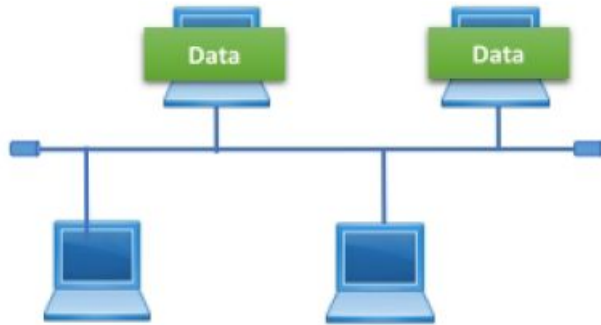
- A **Local Area Network (LAN)** is a privately owned network that operates within a single building like a home, office or factory
- LANs are widely used to connect personal computers and consumer electronics to let them share resources (e.g., printers) and exchange information
- **Wireless LANs** are very popular these days, especially in homes, older office buildings, cafeterias, and other places where it is too much trouble to install cables
- In these systems, every computer has a radio modem and an antenna that it uses to communicate with other computers

More on LAN ...

- In most cases, each computer talks to a device called as **Access Point (AP)**, **wireless router**, or **base station**
- This device relays packets between the wireless computers and also between them and the Internet
- If other computers are close enough, they can communicate directly with one another in a peer-to-peer configuration.
- There is a standard for wireless LANs called **IEEE 802.11**, popularly known as **WiFi**, which has become very widespread

LAN Topology

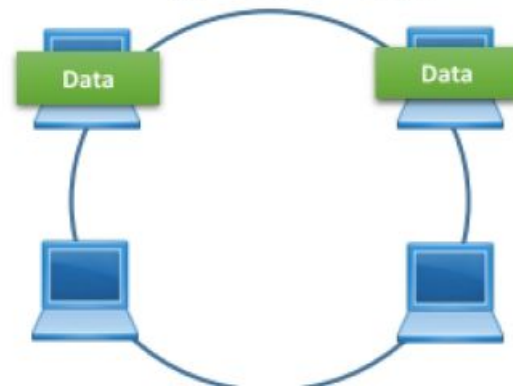
Bus Topology



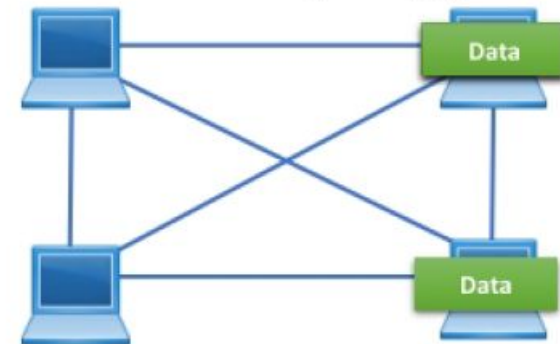
Star Topology



Ring Topology



Mesh Topology



Comparison

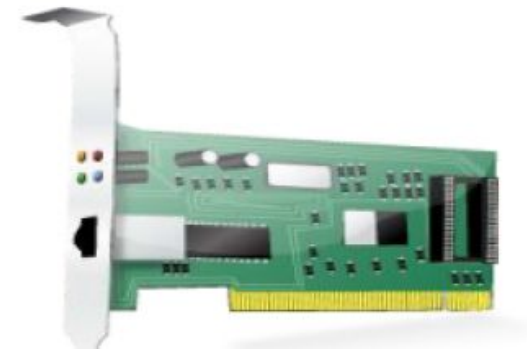
Topology	Advantages	Disadvantages
Star topology	Node Failure doesn't affect the whole network	Coordinator failure cause whole network unable to work
Mesh topology	Highly-Fault Tolerant	Mesh topology is very complex
Cluster Tree Topology	Node Failure doesn't affect the whole network	Beacons must be send to add more nodes
Star-Mesh Topology	Highest Degree of Mobility and Flexibility to rapid changes	It is very complex and takes much effort to handle than any other Topology
Ring Topology	It is very simple compared to other Topologies	Node Failure affects whole network

Setting up a network

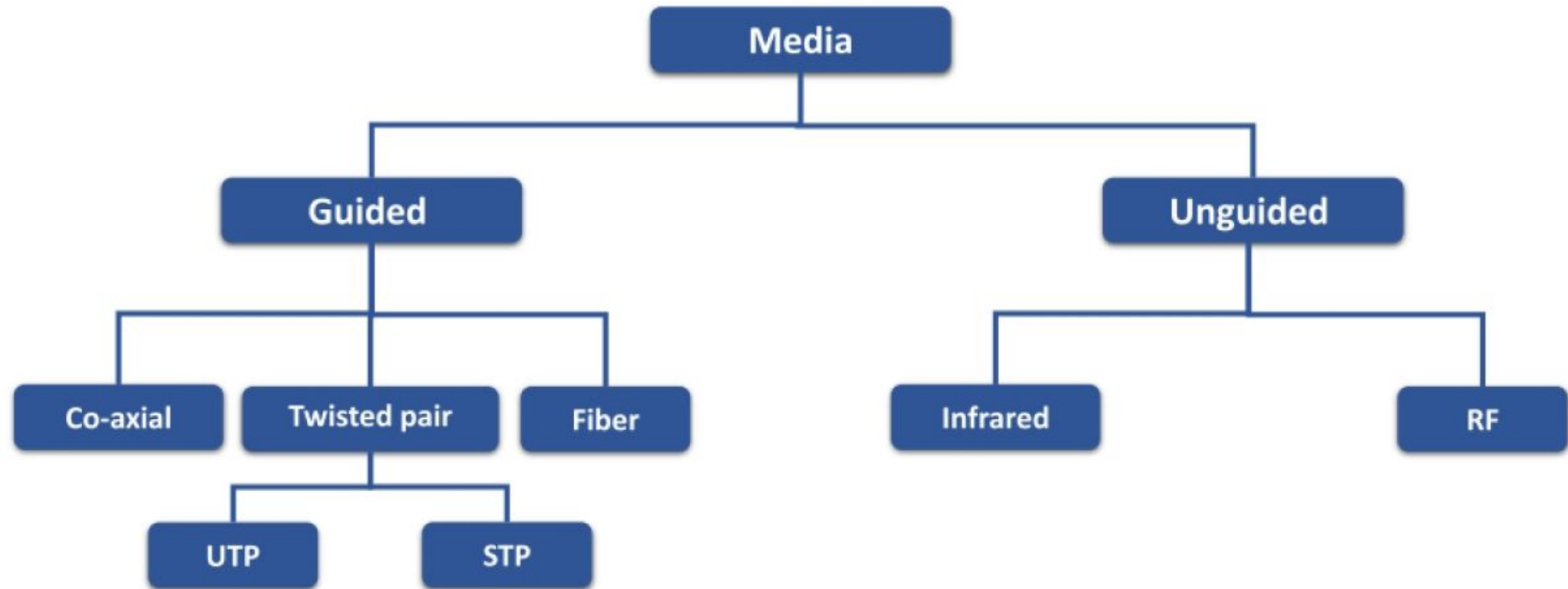
- Network Interface Card (NIC)
- Transmission media
- Network devices

Network Interface Card (NIC)

- NIC is the interface between the computer and the network
- It is also known as the Lan card or Ethernet card
- Ethernet cards have a unique 48 bit address called as MAC (Media access control) address
 - MAC address is also called as Physical address or hardware address
 - The 48 bit MAC address is represented as 12 Hexa-decimal digits
 - Example: 0016.D3FC.603F
- Network cards are available in different speeds
 - Ethernet (10 Mbps)
 - Fast Ethernet (100 Mbps)
 - Gigabit Ethernet (1000 Mbps)



Transmission media



Transmission media



UTP Cable



STP Cable



Co-axial cable



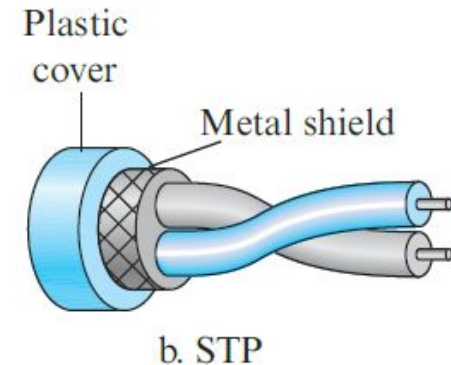
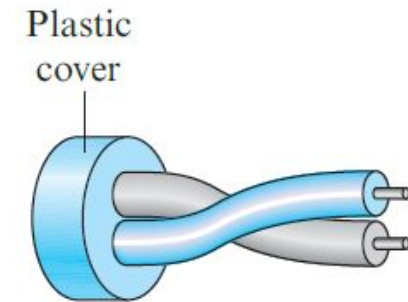
Fiber optic

Transmission Media

- A **transmission medium** can be broadly defined as anything that can carry information from a source to a destination
- For example, the transmission medium for two people having a dinner conversation is the air
- In data communications the definition of the information and the transmission medium is more specific
- Transmission media can be divided into two broad categories:
 - **Guided media (Wired)** include twisted-pair cable (uses electric current), coaxial cable (uses electric current), fiber-optic cable (uses light)
 - **Unguided medium (Wireless)** is free space

Twisted-pair cable

- A twisted pair consists of two conductors (normally copper), each with its own plastic insulation, twisted together
- One of the wires is used to carry signals to the receiver, and the other is used only as a ground reference
- The receiver uses the difference between the two
- In addition to the signal from the sender, interference (noise) and crosstalk may affect both wires and create unwanted signals
- If the two wires are parallel, the effect of these unwanted signals is not the same in both wires because they are at different locations relative to the noise or crosstalk sources
- This results in a difference at the receiver
- By twisting the pairs, a balance is maintained

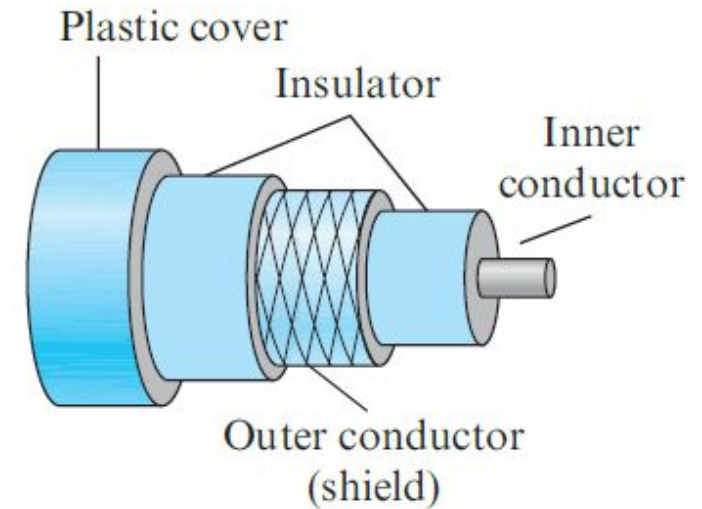


UTP and STP

- The most common twisted-pair cable used in communications is referred to as **unshielded twisted-pair (UTP)**
- IBM has also produced a version of twisted-pair cable for its use called **shielded twisted-pair (STP)**
- STP cable has a metal foil or braided-mesh covering that encases each pair of insulated conductors
- Although metal casing improves the quality of cable by preventing the penetration of noise or crosstalk, it is bulkier and more expensive
- Twisted-pair cables are used in telephone lines to provide voice and data channels
- The local loop—the line that connects subscribers to the central telephone office—commonly uses twisted-pair cables

Coaxial cables

- **Coaxial cable** (or **coax**) carries signals of higher frequency ranges than those in twisted-pair cable
- Instead of having two wires, coax has a central core conductor of solid or stranded wire (usually copper) enclosed in an insulating sheath, which is, in turn, encased in an outer conductor of metal foil, braid, or a combination of the two
- The outer metallic wrapping serves both as a shield against noise and as the second conductor, which completes the circuit
- This outer conductor is also enclosed in an insulating sheath, and the whole cable is protected by a plastic cover

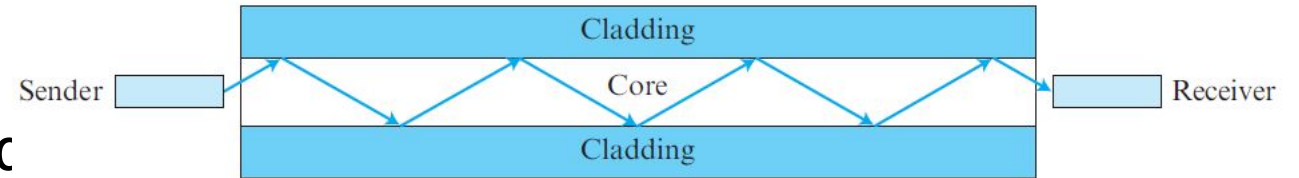


Coaxial cable – Usage

- Coaxial cable was widely used in analog telephone networks where a single coaxial network could carry 10,000 voice signals
 - Later it was used in digital telephone networks where a single coaxial cable could carry digital data up to 600 Mbps
 - However, coaxial cable in telephone networks has largely been replaced today with fiber-optic cable
- Cable TV networks also use coaxial cable
 - In the traditional cable TV network, the entire network used coaxial cable
 - Later, however, cable TV providers replaced most of the media with fiber-optic cable; hybrid networks use coaxial cable only at the network boundaries, near the consumer premises
- Another common application of coaxial cable is in traditional Ethernet LANs
- Because of its high bandwidth, and consequently high data rate, coaxial cable was chosen for digital transmission in early Ethernet LANs

Fiber-optic cable

- A fiber-optic cable is made of glass or plastic and transmits signals in the form of light
- Optical fibers use reflection to guide light through a channel
- A glass or plastic core is surrounded by a cladding of less dense glass or plastic
- The difference in density of the two materials must be such that a beam of light moving through the core is reflected off the cladding instead of being refracted into it

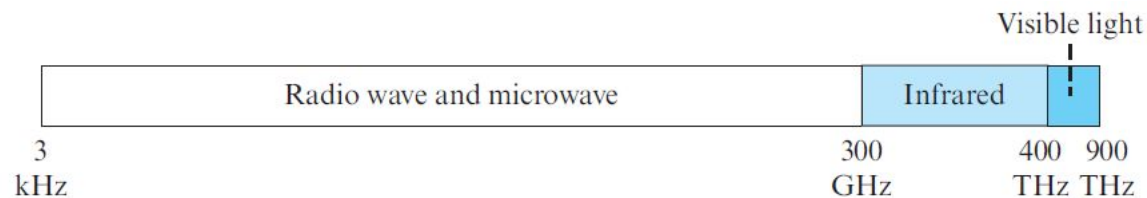


Fiber-optic usage

- Fiber-optic cable is often found in **backbone networks** because its wide bandwidth is cost-effective
- Some cable TV companies use a combination of optical fiber and coaxial cable, thus creating a hybrid network
- Optical fiber provides the backbone structure while coaxial cable provides the connection to the user premises
- This is a cost-effective configuration since the narrow bandwidth requirement at the user end does not justify the use of optical fiber
- Local-area networks such as 100Base-FX network (Fast Ethernet) and 1000Base-X also use fiber-optic cable
- The 10-Gigabit Ethernet operation also uses fiber-optic cable

Unguided media

- Unguided media transport **electromagnetic waves** without using a physical conductor
- This type of communication is often referred to as **wireless communication**
- Signals are normally broadcast through free space and thus are available to anyone who has a device capable of receiving them
- The part of the electromagnetic spectrum, ranging from 3 kHz to 900 THz is used for wireless communication.



Unguided signals

- Unguided signals can travel from source to destination in several ways:
- **Ground propagation:** Radio waves travel through the lowest portion of the atmosphere, hugging the earth. These low-frequency signals emanate in all directions from the transmitting antenna and follow the curvature of the planet. Distance depends on the amount of power in the signal: The greater the power, the greater the distance.
- **Sky propagation:** Higher-frequency radio waves radiate upward into the ionosphere (the layer of atmosphere where particles exist as ions) where they are reflected back to earth. This type of transmission allows for greater distances with lower output power.
- **Line-of-sight propagation:** Very high-frequency signals are transmitted in straight lines directly from antenna to antenna. Antennas must be directional, facing each other, and either tall enough or close enough together not to be affected by the curvature of the earth. Line-of-sight propagation is tricky because radio transmissions cannot be completely focused.

Electromagnetic spectrum

<i>Band</i>	<i>Range</i>	<i>Propagation</i>	<i>Application</i>
VLF (very low frequency)	3–30 kHz	Ground	Long-range radio
LF (low frequency)	30–300 kHz	Ground	Radio beacons
MF (middle frequency)	300 kHz–3 MHz	Sky	AM radio
HF (high frequency)	3–30 MHz	Sky	Citizens band (CB), ship/aircraft communication
VHF (very high frequency)	30–300 MHz	Sky and line-of-sight	VHF TV, FM radio
UHF (ultrahigh frequency)	300 MHz–3 GHz	Line-of-sight	UHF TV, cellular phones, paging, satellite
SHF (superhigh frequency)	3–30 GHz	Line-of-sight	Satellite communication
EHF (extremely high frequency)	30–300 GHz	Line-of-sight	Radar, satellite

Classification of Wireless communication

- Radio waves
- Microwaves
- Infrared waves

Radio waves

- Electromagnetic waves ranging in frequencies between 3 kHz and 1 GHz are normally called **radio waves**; waves ranging in frequencies between 1 and 300 GHz are called **microwaves**
- However, the behavior of the waves, rather than the frequencies, is a better criterion for classification
- Radio waves are omnidirectional
- When an antenna transmits radio waves, they are propagated in all directions. This means that the sending and receiving antennas do not have to be aligned. A sending antenna sends waves that can be received by any receiving antenna.
- The omnidirectional property has a disadvantage, too. The radio waves transmitted by one antenna are susceptible to interference by another antenna that may send signals using the same frequency or band.
- Radio waves, particularly those waves that propagate in the sky mode, can travel long distances
- This makes radio waves a good candidate for long-distance broadcasting such as AM radio
- Radio waves, particularly those of low and medium frequencies, can penetrate walls. This characteristic can be both an advantage and a disadvantage. It is an advantage because, an AM radio can receive signals inside a building. It is a disadvantage because we cannot isolate a communication to just inside or outside a building.
- The radio wave band is relatively narrow, just under 1 GHz, compared to the microwave band

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- The radio wave band is relatively narrow, just under 1 GHz, compared to the microwave band

Microwaves

- Electromagnetic waves having frequencies between 1 and 300 GHz are called **microwaves**
- Microwaves are unidirectional. When an antenna transmits microwaves, they can be narrowly focused. This means that the sending and receiving antennas need to be aligned. The unidirectional property has an obvious advantage. A pair of antennas can be aligned without interfering with another pair of aligned antennas.
- Microwave propagation is line-of-sight. Since the towers with the mounted antennas need to be in direct sight of each other, towers that are far apart need to be very tall.
- The curvature of the earth as well as other blocking obstacles do not allow two short towers to communicate by using microwaves.
- Repeaters are often needed for long distance communication.
- Very high-frequency microwaves cannot penetrate walls. This characteristic can be a disadvantage if receivers are inside buildings.
- Microwaves, due to their unidirectional properties, are very useful when **unicast** (one-to-one) communication is needed between the sender and the receiver. They are used in cellular phones, satellite networks, and wireless LANs.

Infrared

- **Infrared waves**, with frequencies from 300 GHz to 400 THz (wavelengths from 1 mm to 770 nm), can be used for short-range communication
- Infrared waves, having high frequencies, cannot penetrate walls. This advantage prevents interference between one system and another; a short-range communication system in one room cannot be affected by another system in the next room.
- When we use our infrared remote control, we do not interfere with the use of the remote by our neighbors
- However, this same characteristic makes infrared signals useless for long-range communication
- In addition, we cannot use infrared waves outside a building because the sun's rays contain infrared waves that can interfere with the communication

Wired LAN - Ethernet

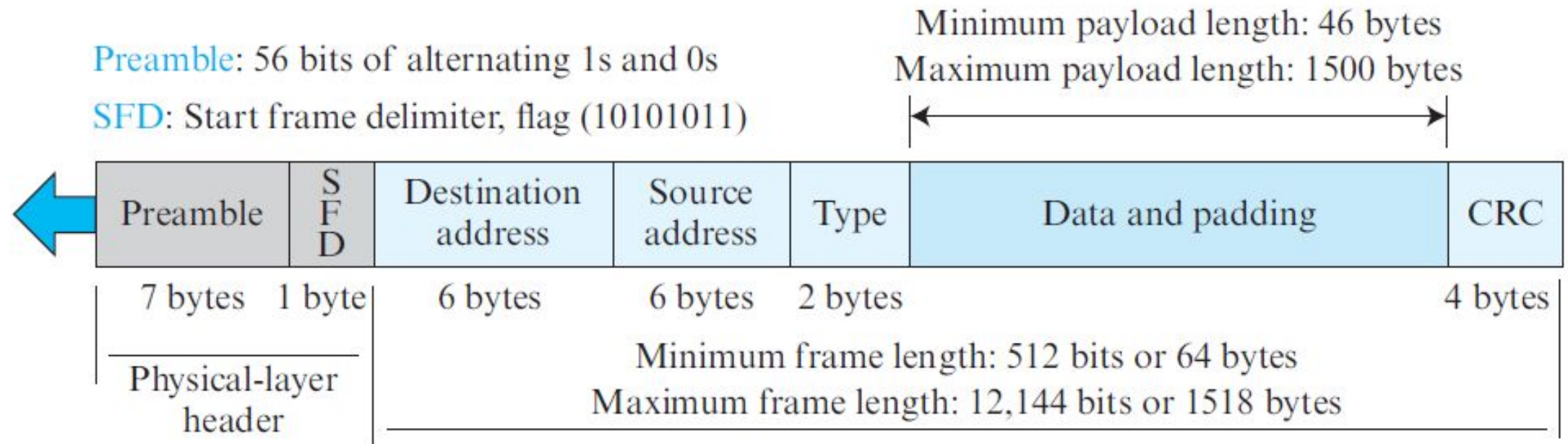
What is Ethernet?

- In the 1980s and 1990s several different types of LANs were used: Ethernet, Token Ring, Token Bus, and FDDI (Fiber Distribution Data Interface)
- Almost every LAN except Ethernet has disappeared from the marketplace because Ethernet was able to update itself to meet the needs of the time. Main reason for its success: it could evolve with the demand for higher transmission rates.
- The Ethernet LAN was developed in the 1970s by Robert Metcalfe and David Boggs.
- Since then, it has gone through four generations: **Standard Ethernet (10 Mbps), Fast Ethernet (100 Mbps), Gigabit Ethernet (1 Gbps), and 10 Gigabit Ethernet (10 Gbps).**

IEEE 802

- In 1985, the Computer Society of the IEEE started a project, called **Project 802**, to set standards to enable intercommunication among equipment from a variety of manufacturers.
- Project 802 does not seek to replace any part of the OSI model or TCP/IP protocol suite.
- Instead, it is a way of specifying functions of the physical layer and the data-link layer of major LAN protocols.

Ethernet frame format



Connectionless and unreliable service

- Ethernet provides a connectionless service, which means each frame sent is independent of the previous or next frame.
- Ethernet has no connection establishment or connection termination phases.
- The sender sends a frame whenever it has it; the receiver may or may not be ready for it.
- The sender may overwhelm the receiver with frames, which may result in dropping frames.
- If a frame drops, the sender will not know about it.

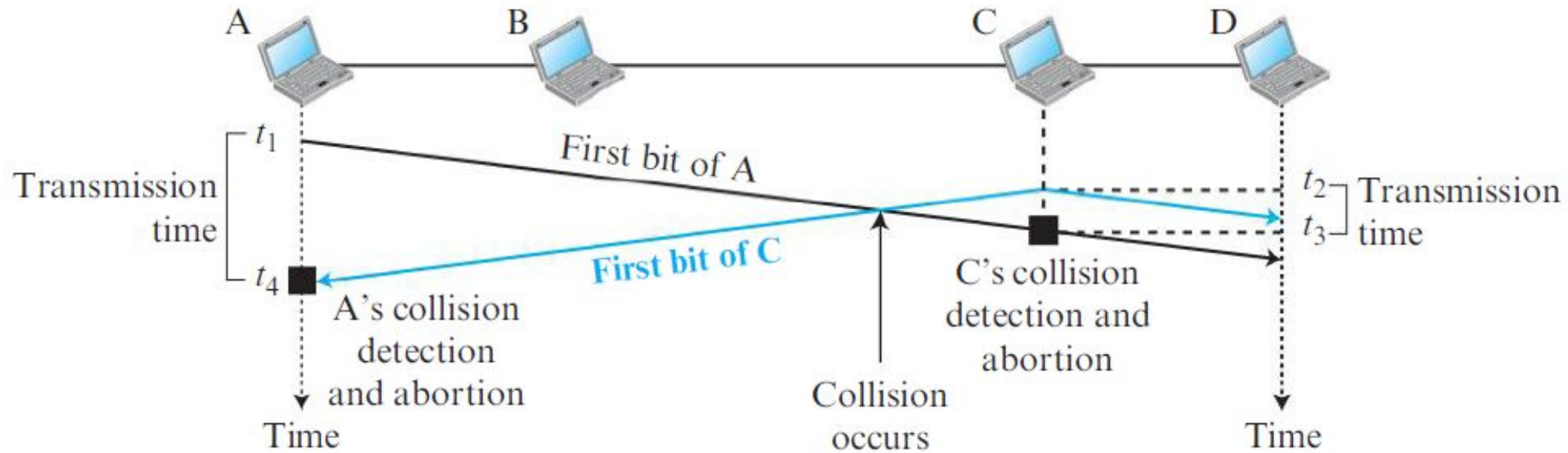
Addressing

- Each station on an Ethernet network (such as a PC, workstation, or printer) has its own **network interface card (NIC)**.
- The NIC fits inside the station and provides the station with a link-layer address.
- The Ethernet address is 6 bytes (48 bits), normally written in hexadecimal notation, with a colon between the bytes.
- For example, the following shows an Ethernet MAC address:
- **E4-46-B0-03-74-D2**

CSMA/CD

- **Carrier sense multiple access with collision detection (CSMA/CD)** augments the algorithm to handle the collision.
- In this method, a station monitors the medium after it sends a frame to see if the transmission was successful. If so, the station's job is finished.
- If, however, there is a collision, the frame is sent again.
- To better understand CSMA/CD, let us look at the first bits transmitted by the two stations involved in the collision. Although each station continues to send bits in the frame until it detects the collision, we show what happens as the first bits collide.

CSMA/CD



Modern Ethernet

- Two kinds of Ethernet exist: **classic Ethernet**, which solves the multiple access problem using the techniques we have studied earlier; and **switched Ethernet**, in which devices called **switches** are used to connect different computers.
- It is important to note that, while they are both referred to as Ethernet, they are quite different.
- Classic Ethernet is the original form and ran at rates from 3 to 10 Mbps.
- Switched Ethernet is what Ethernet has become and runs at 100, 1000, and 10,000 Mbps, in forms called **fast Ethernet**, **gigabit Ethernet**, and **10 gigabit Ethernet**.
- In practice, only switched Ethernet is used nowadays.

Switched Ethernet example: Gigabit ethernet

- The need for an even higher data rate resulted in the design of the **Gigabit Ethernet Protocol** (1000 Mbps).
- The IEEE committee calls the Standard 802.3z.
- The goals of the Gigabit Ethernet were to upgrade the data rate to 1 Gbps, but keep the address length, the frame format, and the maximum and minimum frame length the same.
- There is a central switch connected to all computers or other switches.
- In this mode, for each input port, each switch has buffers in which data are stored until they are transmitted.
- Since the switch uses the destination address of the frame and sends a frame out of the port connected to that particular destination, there is no collision. This means that CSMA/CD is not used.
- Lack of collision implies that the maximum length of the cable is determined by the signal attenuation in the cable, not by the collision detection process.

Switched Ethernet: Using hub or switch

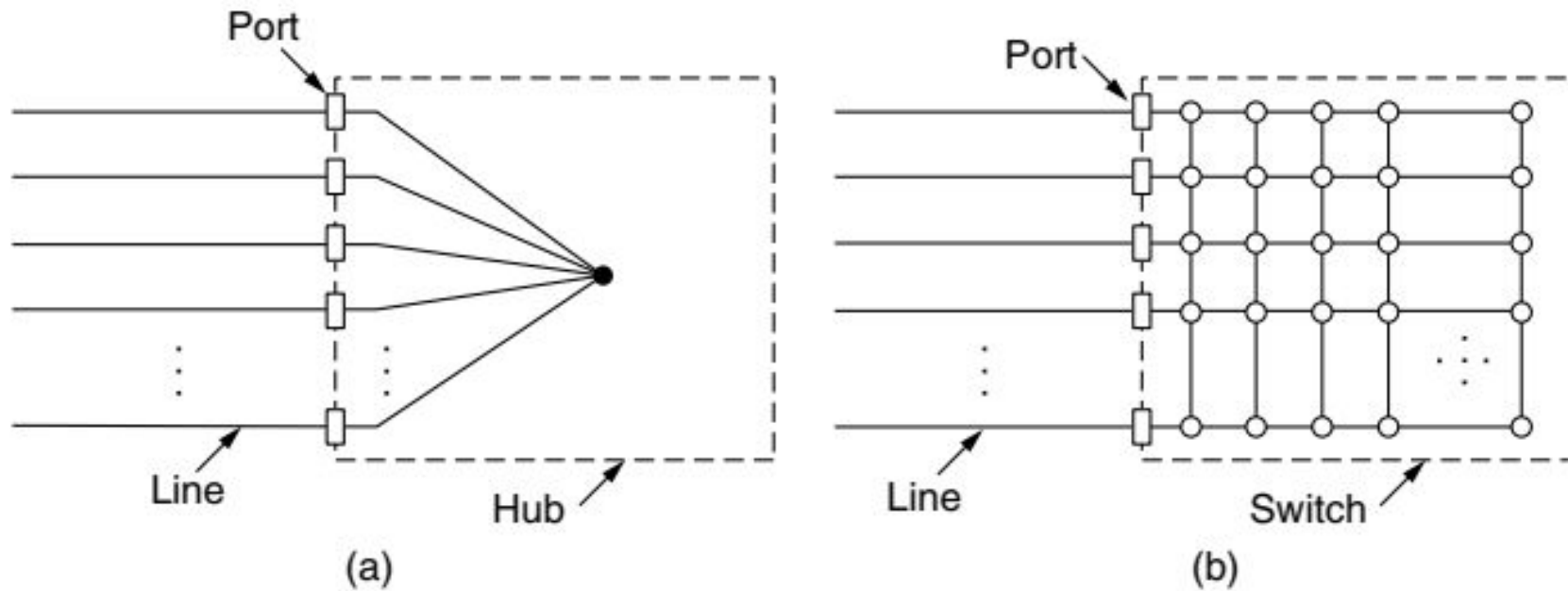


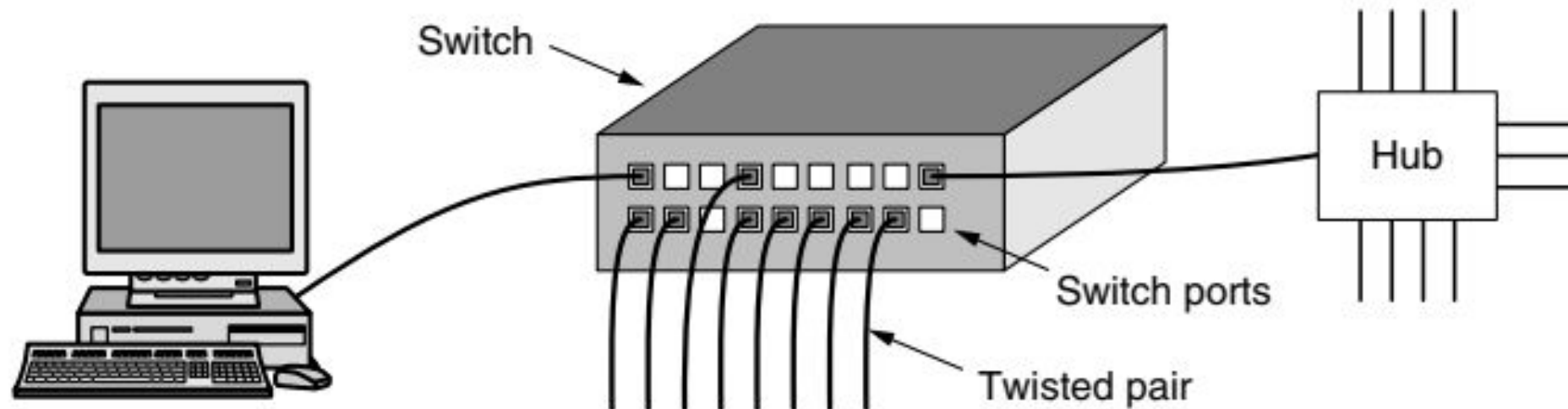
Figure 17. (a) Hub. (b) Switch.

Switched Ethernet - Hub

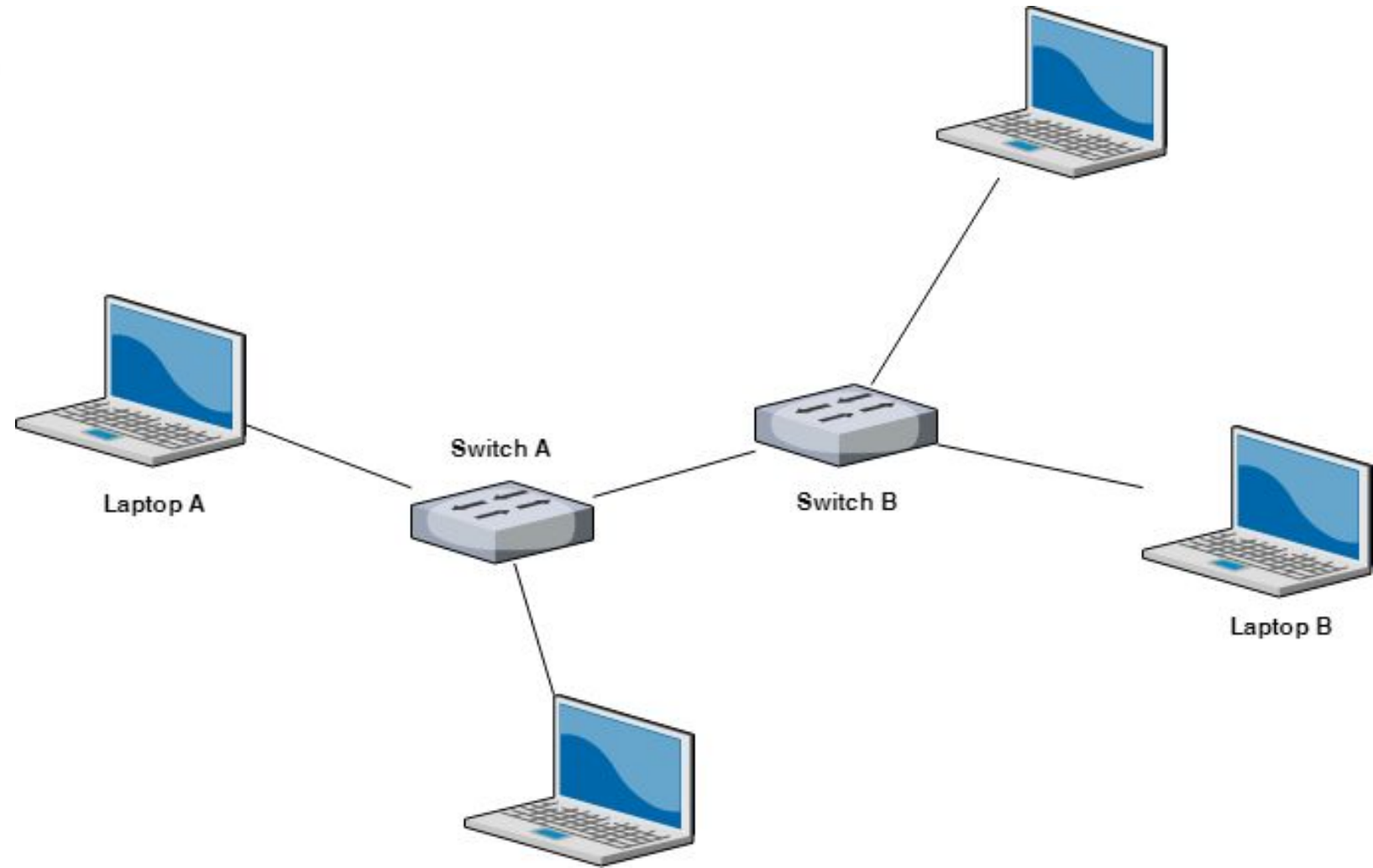
- This uses a different kind of wiring pattern, in which each station has a dedicated cable running to a central **hub**.
- Hub is a dumb device, works only at the physical layer.
- Switch is intelligent, can filter frames, like bridge.
- Earlier, switches were expensive. However, now we rarely see hubs. Switches are used almost everywhere.

Switched Ethernet - Switch

- From the outside, a **switch** looks just like a hub.
- They are both boxes, typically with 4 to 48 ports, each with a standard RJ-45 connector for a twisted-pair cable.
- If the switch itself fails, it will need to be replaced.



Multiple switches are used these days (Not hubs) ...



Use of switch

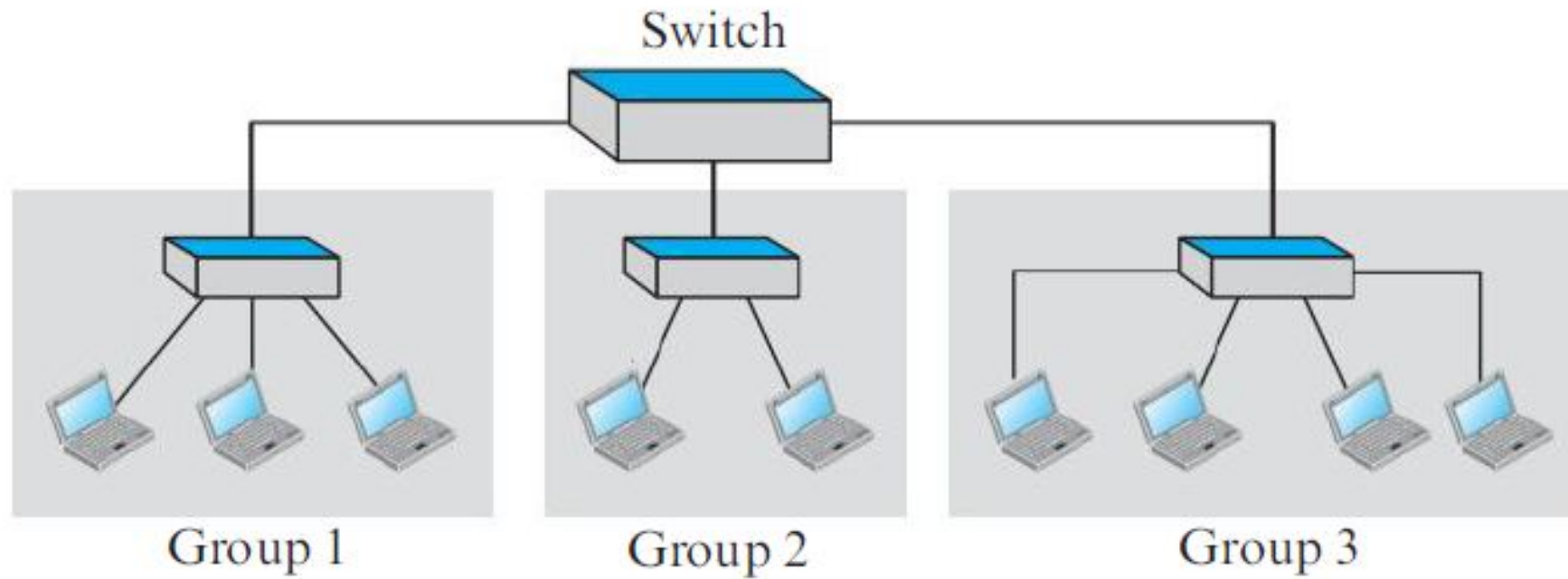
- Switches only output frames to the ports for which those frames are destined.
- When a switch port receives an Ethernet frame from a station, the switch checks the Ethernet addresses to see which port the frame is destined for.
- This step requires the switch to be able to work out which ports correspond to which addresses.
- The switch then forwards the frame over its high-speed backplane to the destination port.
- The destination port then transmits the frame on the wire so that it reaches the intended station. None of the other ports even knows the frame exists.

Virtual LAN (VLAN)

VLAN Basics

- A station/host is considered part of a LAN if it physically belongs to that LAN.
- The criterion of membership is geographic.
- What happens if we need a virtual connection between two stations belonging to two different physical LANs?
- We can roughly define a **virtual local area network (VLAN)** as a local area network configured by software, not by physical wiring.
- Consider a LAN in which ten stations are grouped into three LANs that are connected by a switch. (Figure on the next slide)

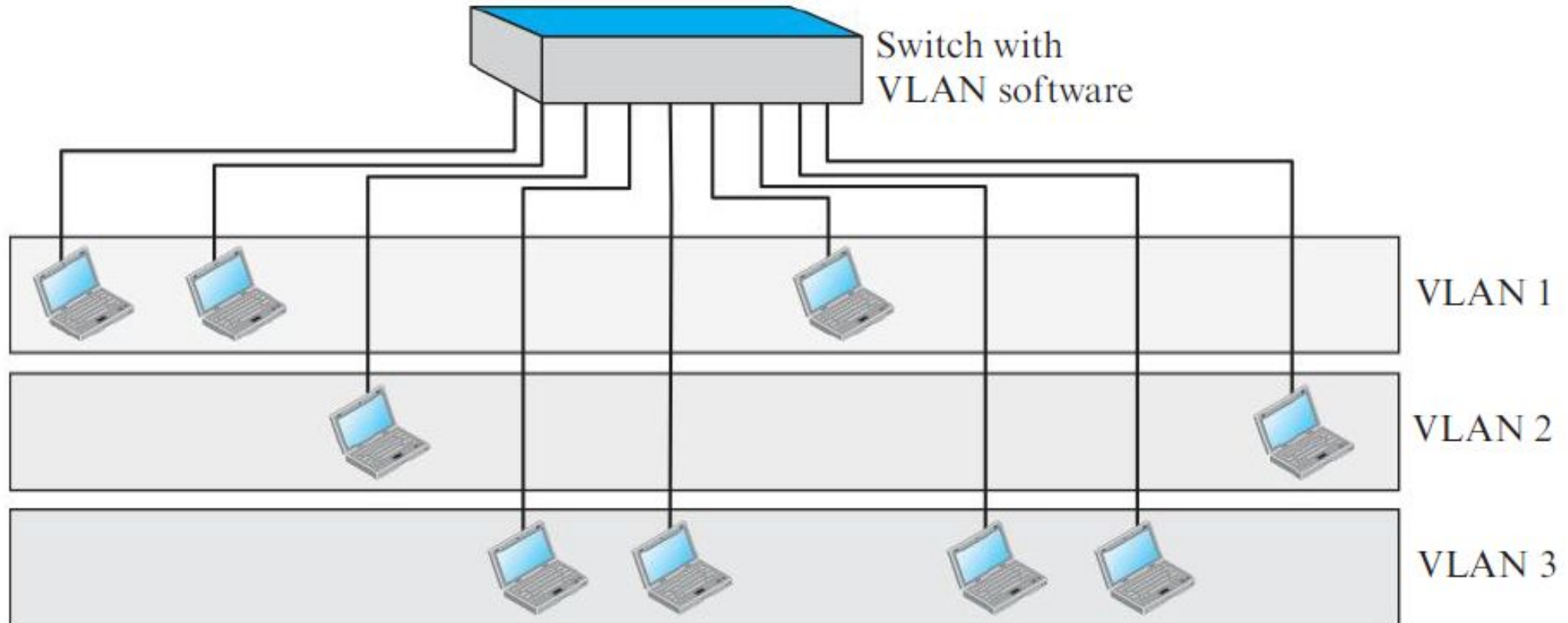
Sample LAN



VLAN – The need

- The first three engineers work together as the first group, the next two engineers work together as the second group, and the last four engineers work together as the third group.
- The LAN is configured to allow this arrangement.
- But what would happen if the administrators needed to move two engineers from the first group to the third group, to speed up the project being done by the third group?
- The LAN configuration would need to be changed.
- The network technician must rewire.
- The problem is repeated if, in another week, the two engineers move back to their previous group.
- In a switched LAN, changes in the work group mean physical changes in the network configuration.
- Solution: VLAN

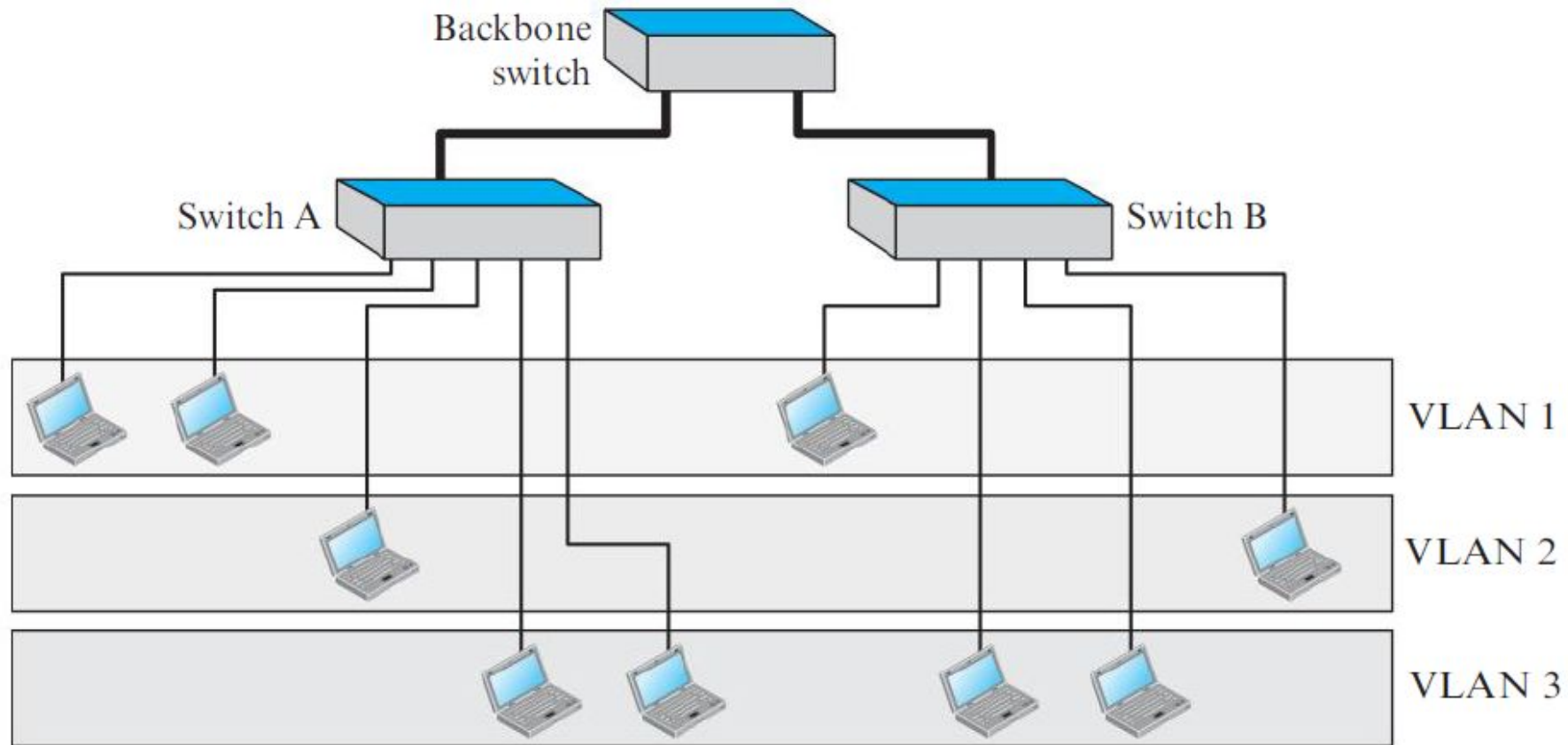
VLAN



How it works?

- Each VLAN is a work group in the organization. If a person moves from one group to another, there is no need to change the physical configuration. The group membership in VLANs is defined by software, not hardware. Any station can be logically moved to another VLAN.
- All members belonging to a VLAN can receive broadcast messages sent to that particular VLAN. This means if a station moves from VLAN 1 to VLAN 2, it receives broadcast messages sent to VLAN 2, but no longer receives broadcast messages sent to VLAN 1.
- It is obvious that the problem in our previous example can easily be solved by using VLANs. Moving engineers from one group to another through software is easier than changing the configuration of the physical network.
- VLAN technology even allows the grouping of stations connected to different switches in a VLAN. (See figure on the next slide)

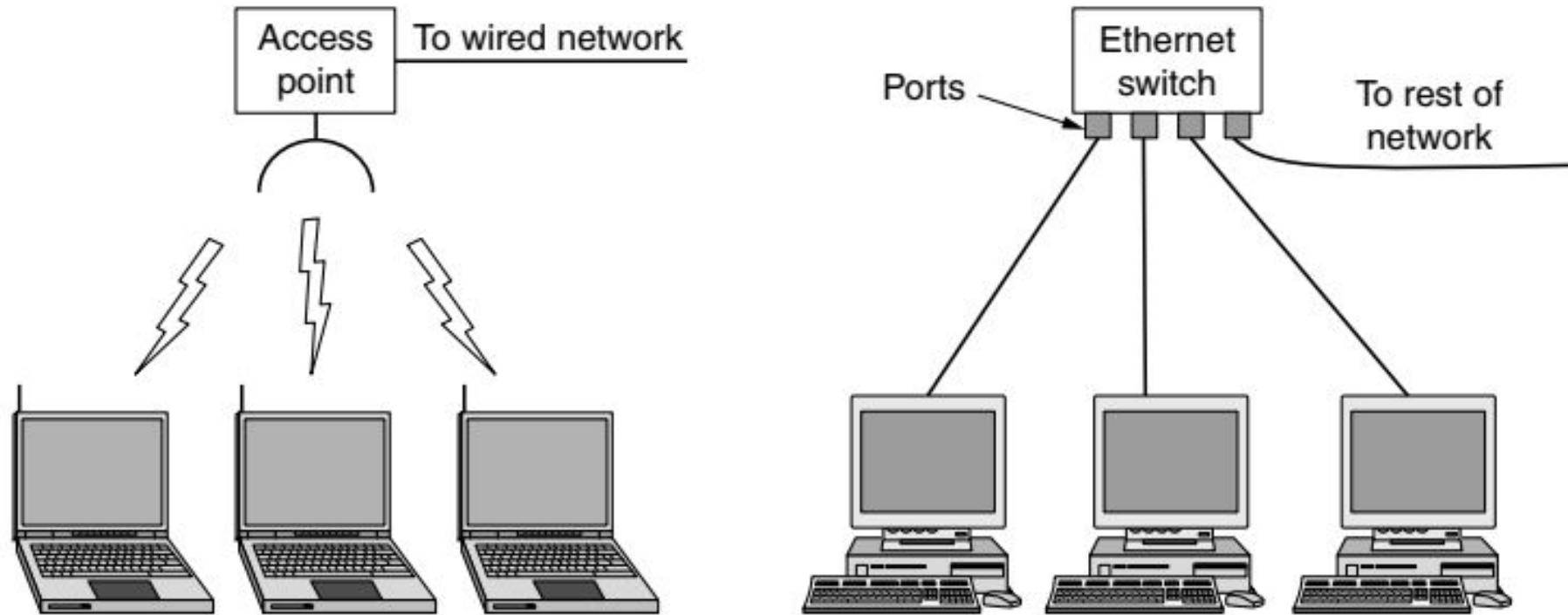
Two switches with a backbone using VLAN



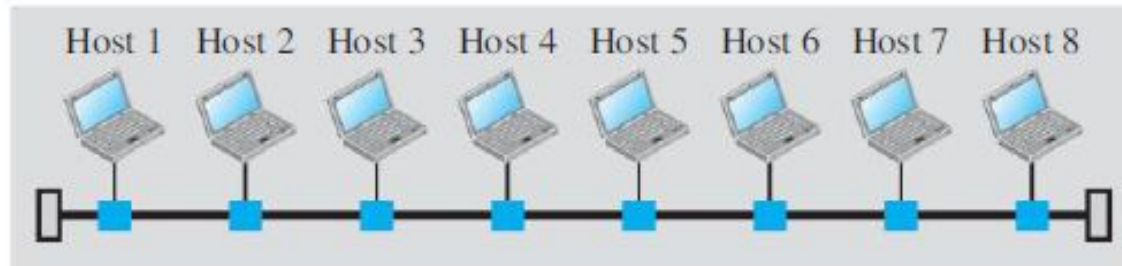
Where can it be used?

- This is a good configuration for a company with two separate buildings.
- Each building can have its own switched LAN connected by a backbone.
- People in the first building and people in the second building can be in the same work group even though they are connected to different physical LANs.

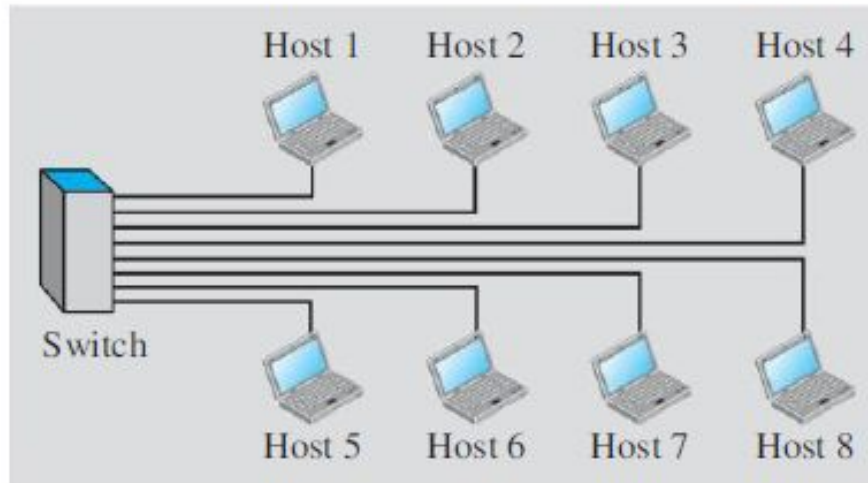
Wired and Wireless LAN



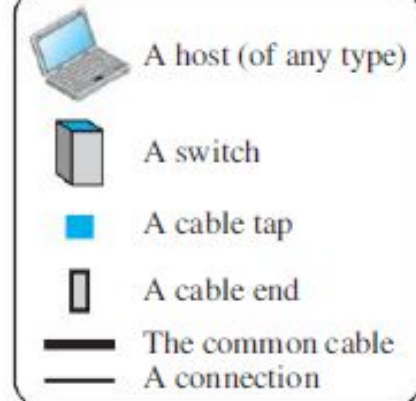
LAN – Earlier and Now



a. LAN with a common cable (past)



Legend



LAN Transmission Technologies

- Wired LANs use a range of different transmission technologies
- Most of them use copper wires, but some use optical fiber
- LANs are restricted in size, which means that the worst-case transmission time is bounded and known in advance
- Knowing these bounds helps with the task of designing network protocols
- Typically, wired LANs run at speeds of 100 Mbps to 1 Gbps, have low delay (microseconds or nanoseconds), and make very few errors
- Newer LANs can operate at up to 10 Gbps
- In contrast, WiFi runs at speeds anywhere from 11 to hundreds of Mbps
- Wired is always faster and more reliable

Wireless LAN

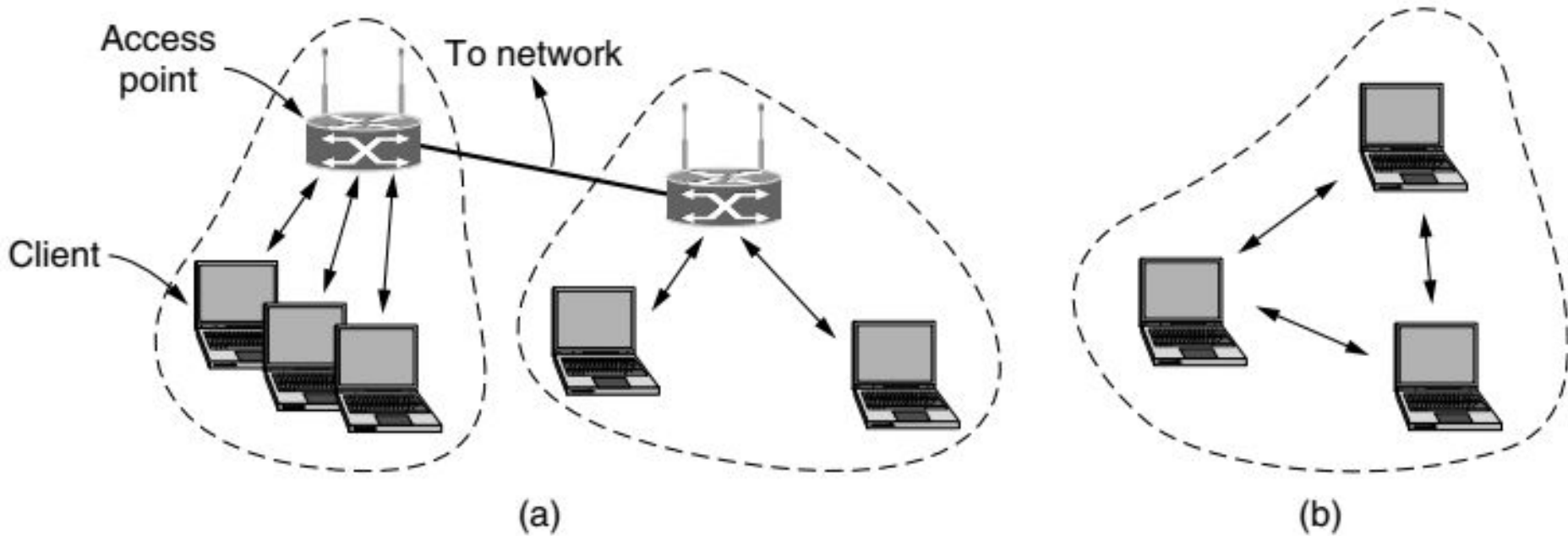


Figure 23. 802.11 architecture. (a) Infrastructure mode. (b) Ad-hoc mode.

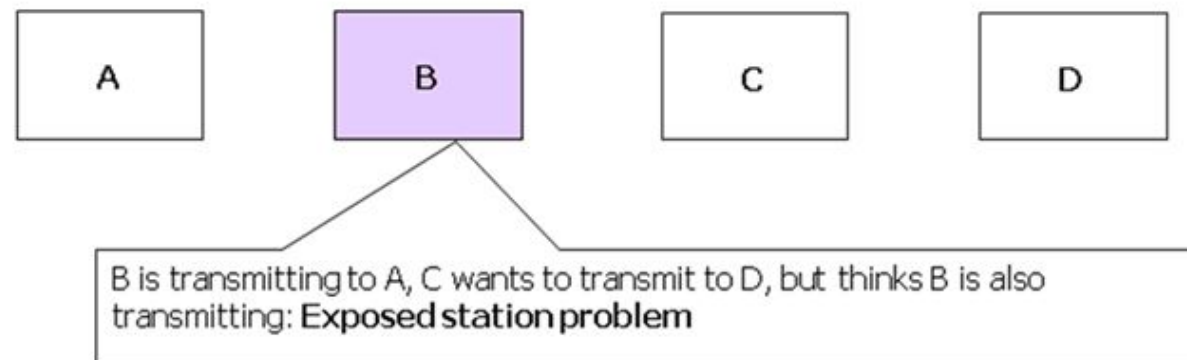
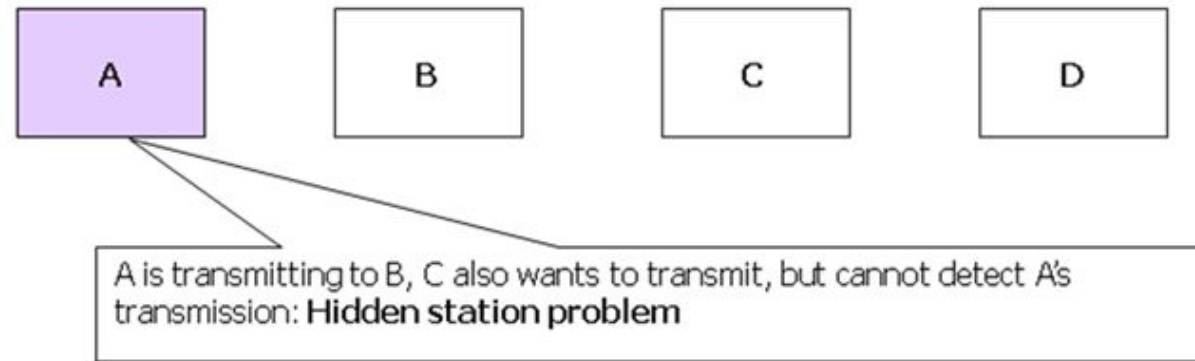
Wireless LAN – Infrastructure mode

- **802.11** networks (**WiFi**) can be used in two modes. The most popular mode is to connect clients such as laptops and smartphones to another network, such as the office/college network or the Internet
- This is called as **infrastructure mode**
- Each client is associated with an **AP (Access Point)** that is in turn connected to the other network
- The client sends and receives its packets via the AP
- Several access points may be connected together, typically by a wired network called a **distribution system**, to form an extended 802.11 network

Wireless LAN – Ad hoc mode

- In the **ad hoc network** mode, we have a collection of computers that can directly send frames to each other
- There is no access point
- Since the Internet is the most popular application, ad hoc networks are not very popular

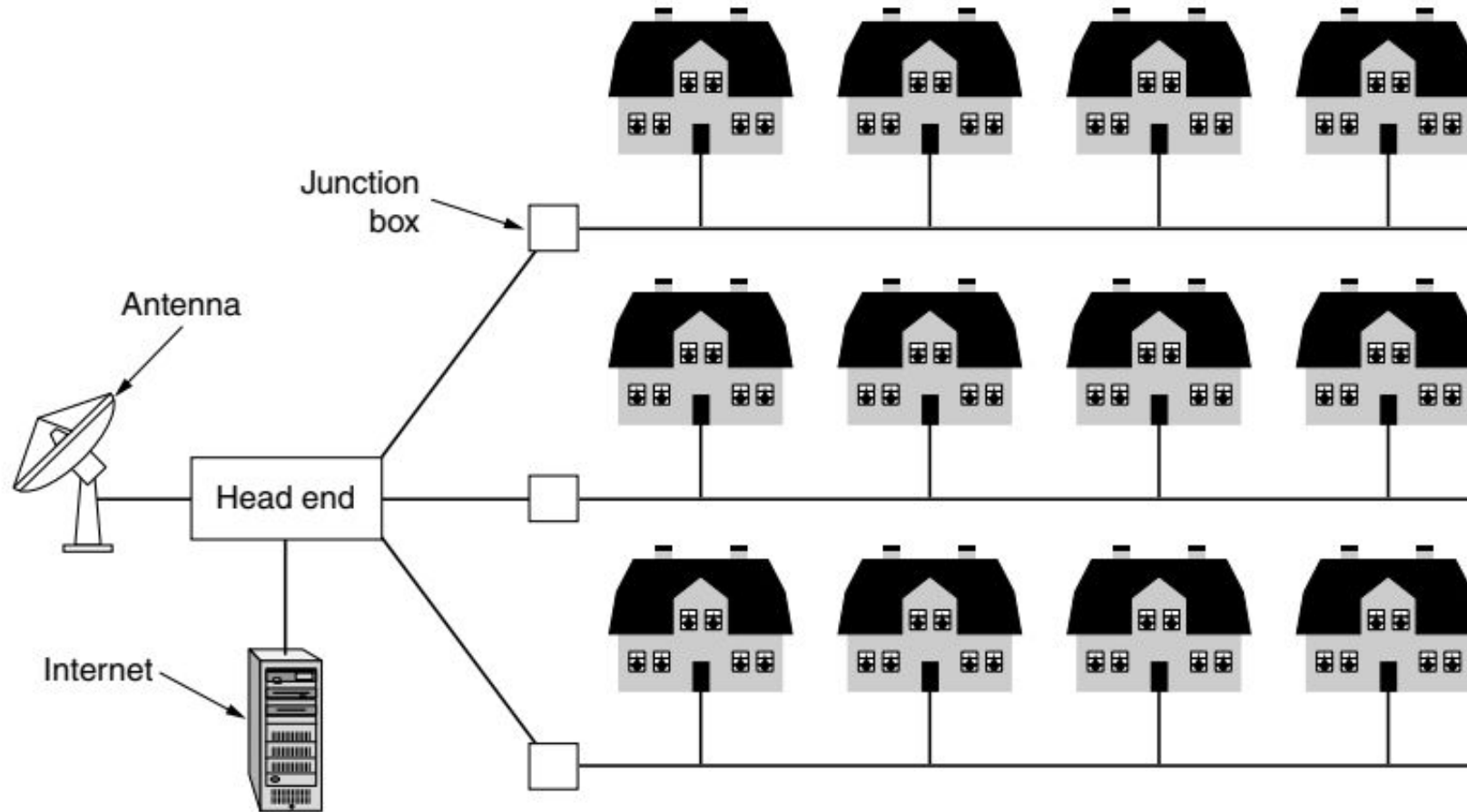
Hidden and Exposed Station Problems



Metropolitan Area Network (MAN)

- A MAN (Metropolitan Area Network) covers a city
- The best-known examples of MANs are the cable television networks available in many cities
- See figure

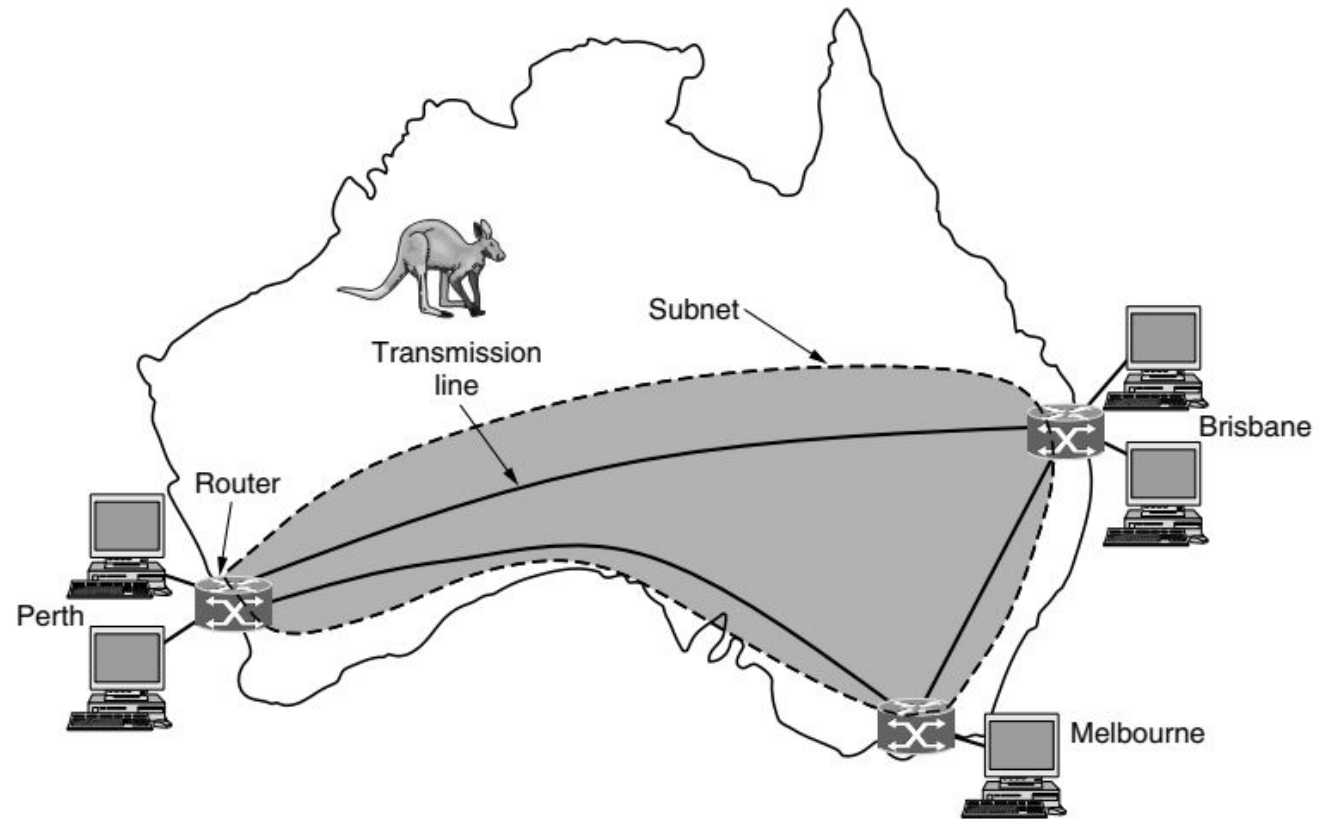
MAN



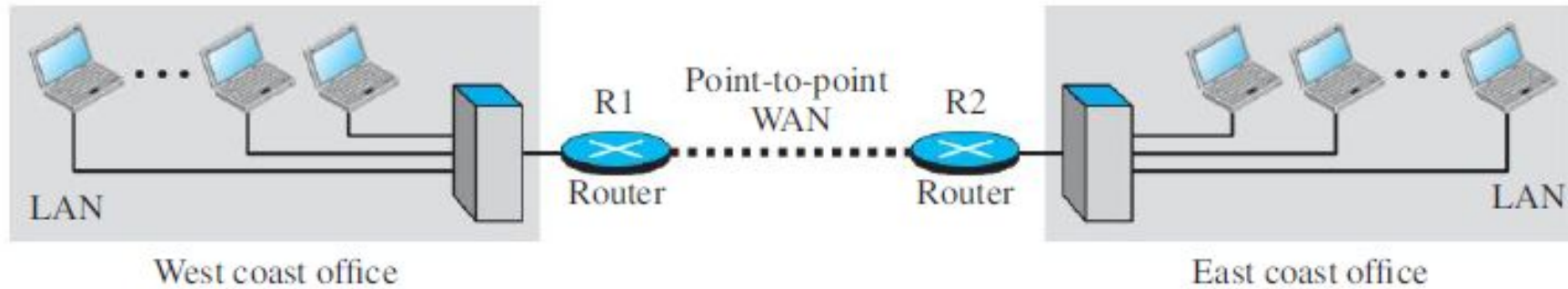
Wide Area Network (WAN)

- A **Wide Area Network (WAN)** spans a large geographical area, often a country or continent
- In most WANs, the network consists of two distinct components: **transmission lines** and **switching elements**
- Transmission lines move bits between machines. They can be made of copper wire, optical fiber, or even radio links. Most companies do not have their own transmission lines, so instead they lease the lines from a telecommunications company.
- Switching elements, or just switches, are specialized computers that connect two or more transmission lines. When data arrive on an incoming line, the switching element must choose an outgoing line on which to forward them. These switching computers have been called by various names in the past; the name **router** is now most commonly used. (See figure)

WAN



WAN Connecting LANs



Internetwork

- Many networks exist in the world, often with different hardware and software
- People connected to one network often want to communicate with people attached to a different one
- The fulfillment of this desire requires that different, and frequently incompatible, networks be connected
- A collection of interconnected networks is called an **internetwork** or **internet**
- The general name for a machine that makes a connection between two or more networks and provides the necessary translation, both in terms of hardware and software, is a **gateway**

Protocol

- Communication between two people or two devices needs to follow some **protocol**
- A protocol is a set of rules that governs communication
- For example, in a face-to-face communication between two persons, there is a set of implicit rules in each culture that define how two persons should start the communication, how to continue the communication, and how to end the communication
- Similarly, in a telephone conversation, there are a set of rules that we need to follow

Network Software

- The first computer networks were designed with the hardware as the main concern and the software as an afterthought
- This strategy no longer works
- Network software is now highly structured using a **protocol stack**
- To reduce their design complexity, most networks are organized as a stack of **layers** or **levels**, each one built upon the one below it
- The number of layers, the name of each layer, the contents of each layer, and the function of each layer differ from network to network
- The purpose of each layer is to offer certain services to the higher layers while shielding those layers from the details of how the offered services are actually implemented
- In a sense, each layer is a kind of virtual machine, offering certain services to the layer above it

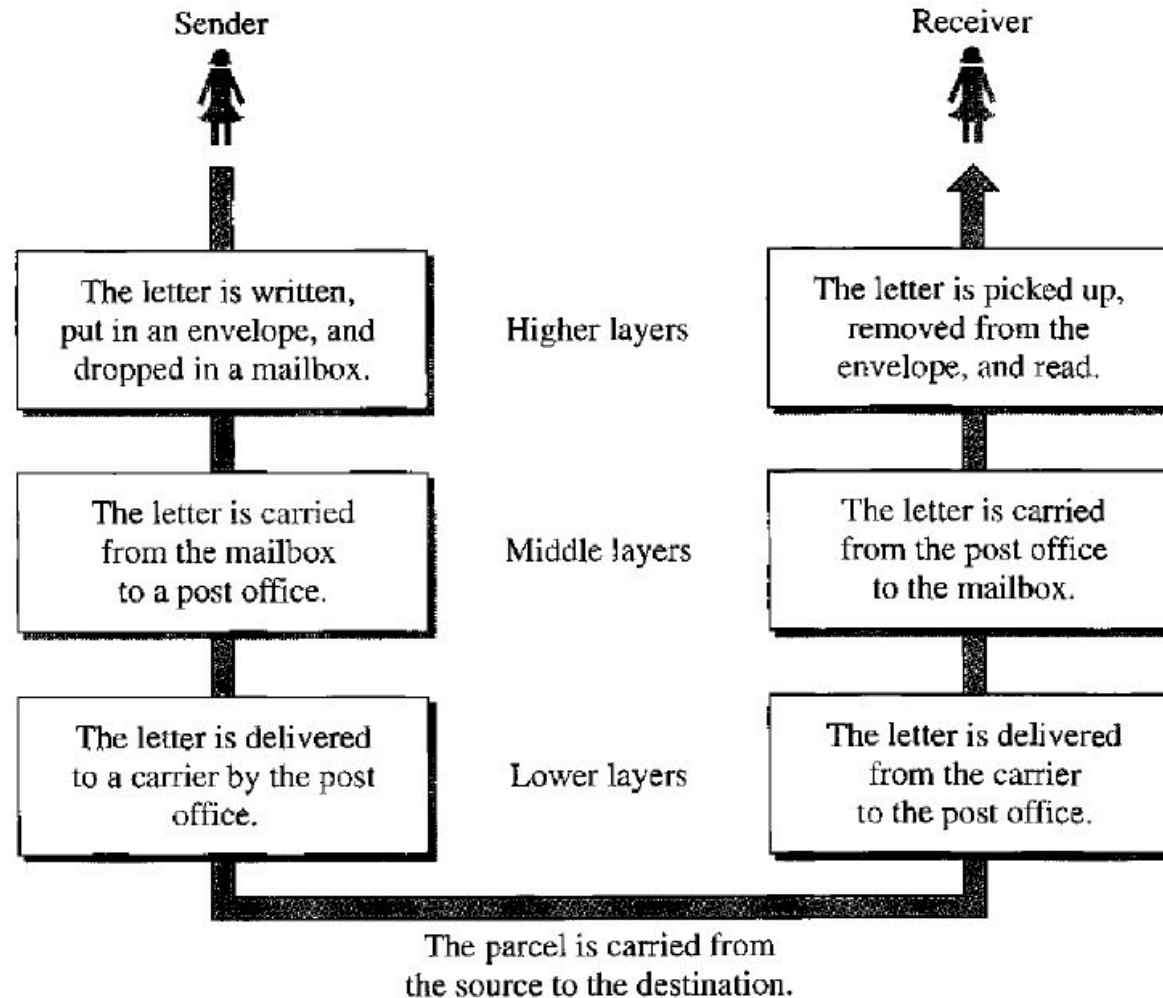
Protocol Hierarchies

- When layer n on one machine carries on a conversation with layer n on another machine, the rules and conventions used in this conversation are collectively known as the layer n **protocol**
- Basically, a protocol is an agreement between the communicating parties on how communication is to proceed
- In reality, no data are directly transferred from layer n on one machine to layer n on another machine
- Instead, each layer passes data and control information to the layer immediately below it, until the lowest layer is reached
- Below layer 1 is the physical medium through which actual communication occurs
- In the figure (on one of the next slides), virtual communication is shown by dotted lines and physical communication by solid lines

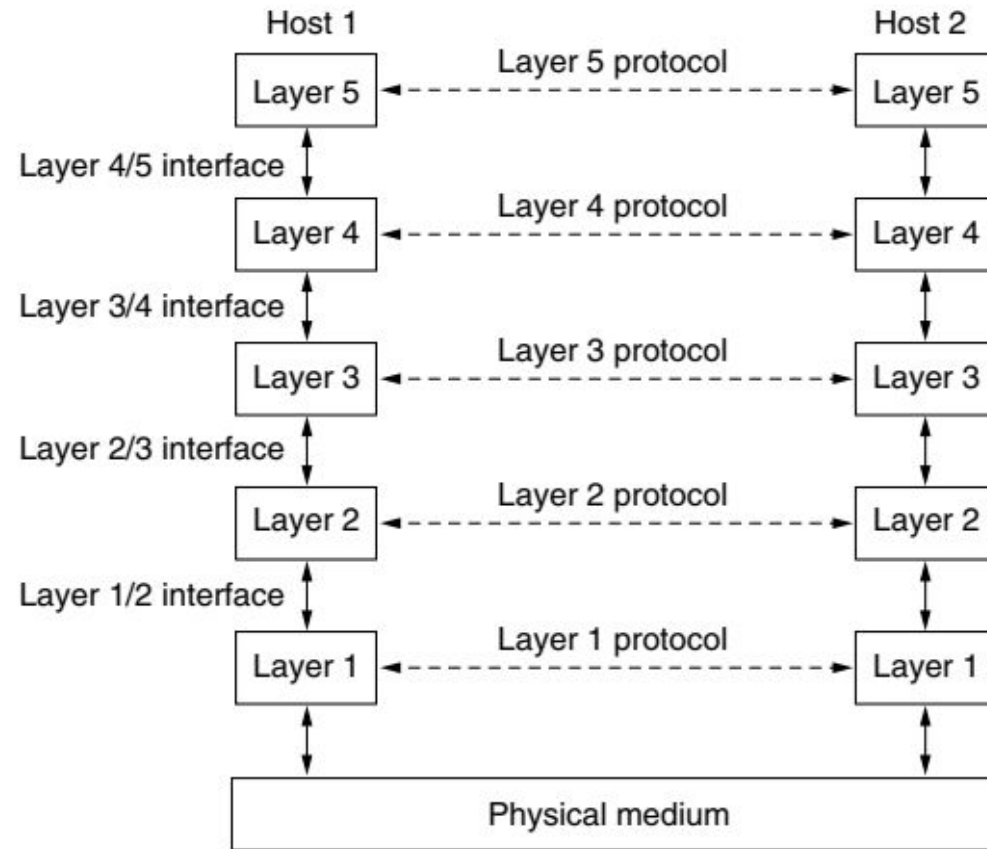
Protocol Hierarchies

- Between each pair of adjacent layers is an **interface**
- The interface defines which primitive operations and services the lower layer makes available to the upper one
- When network designers decide how many layers to include in a network and what each one should do, one of the most important considerations is defining clean interfaces between the layers
- Doing so, in turn, requires that each layer perform a specific collection of well-understood functions
- In addition to minimizing the amount of information that must be passed between layers, clearcut interfaces also make it simpler to replace one layer with a completely different protocol or implementation (e.g., replacing all the telephone lines by satellite channels) because all that is required of the new protocol or implementation is that it offer exactly the same set of services to its upstairs neighbor as the old one did

Protocol Hierarchy Example



Layers, Protocols, Interfaces



Design Issues for Layers – Errors

- Reliability is the design issue of making a network that operates correctly even though it is made up of a collection of components that are themselves unreliable
- Think about the bits of a packet traveling through the network
- There is a chance that some of these bits will be received damaged (0 to 1 or 1 to 0) due to electrical noise, random wireless signals, hardware flaws, software bugs and so on
- How is it possible that we find and fix these errors?
- One mechanism for finding errors in received information uses codes for **error detection**
- Information that is incorrectly received can then be retransmitted until it is received correctly
- More powerful codes allow for **error correction**, where the correct message is recovered from the possibly incorrect bits that were originally received
- Both of these mechanisms work by adding redundant information

Design Issues for Layers – Routing

- Another reliability issue is finding a working path through a network
- Often there are multiple paths between a source and destination, and in a large network, there may be some links or routers that are broken
- Suppose that the network is down in Germany
- Packets sent from London to Rome via Germany will not get through, but we could instead send packets from London to Rome via Paris
- The network should automatically make this decision
- This topic is called **routing**

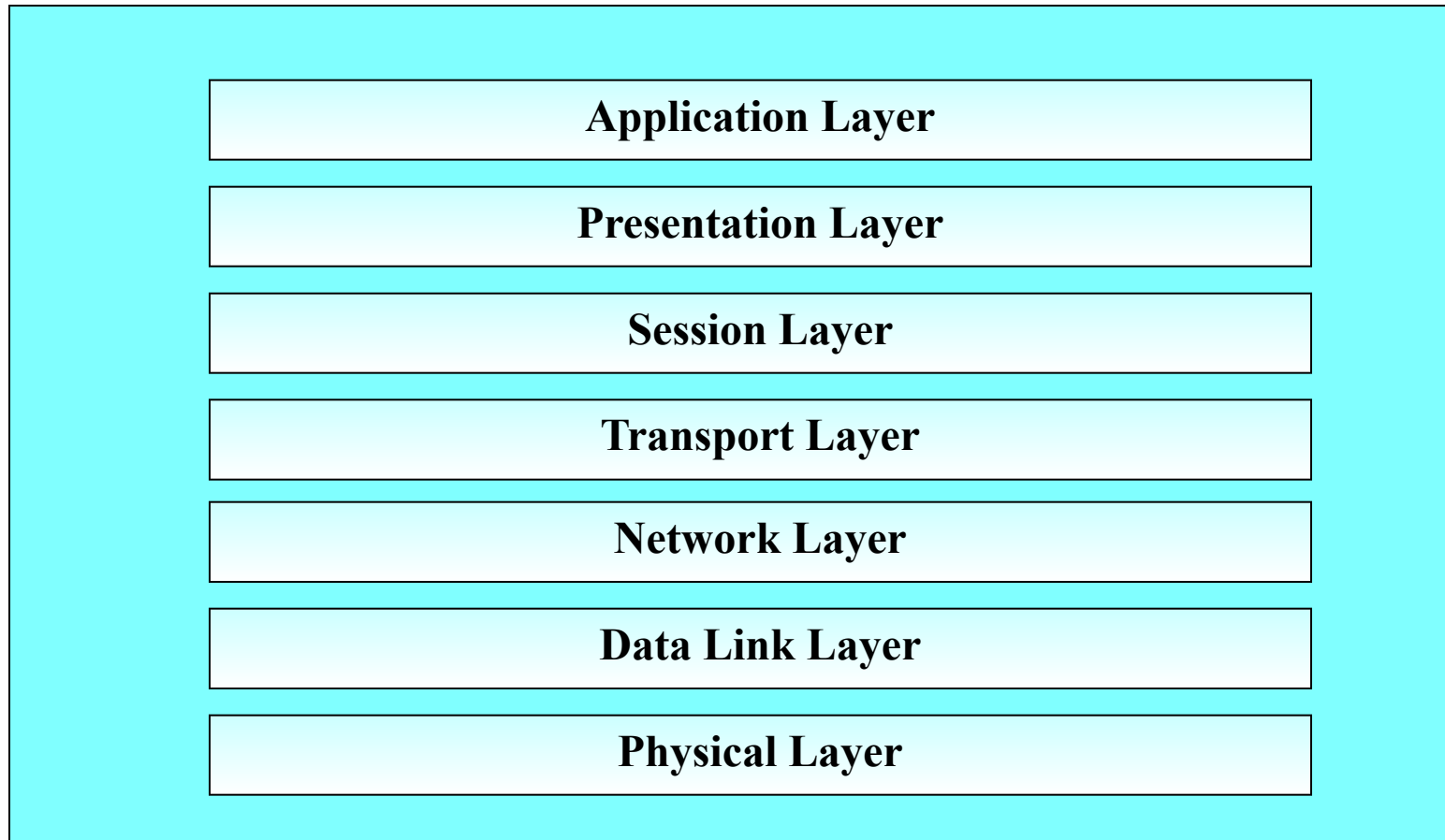
Design Issues for Layers – Addressing

- Since there are many computers on the network, every layer needs a mechanism for identifying the senders and receivers that are involved in a particular message
- This mechanism is called **addressing** or **naming**

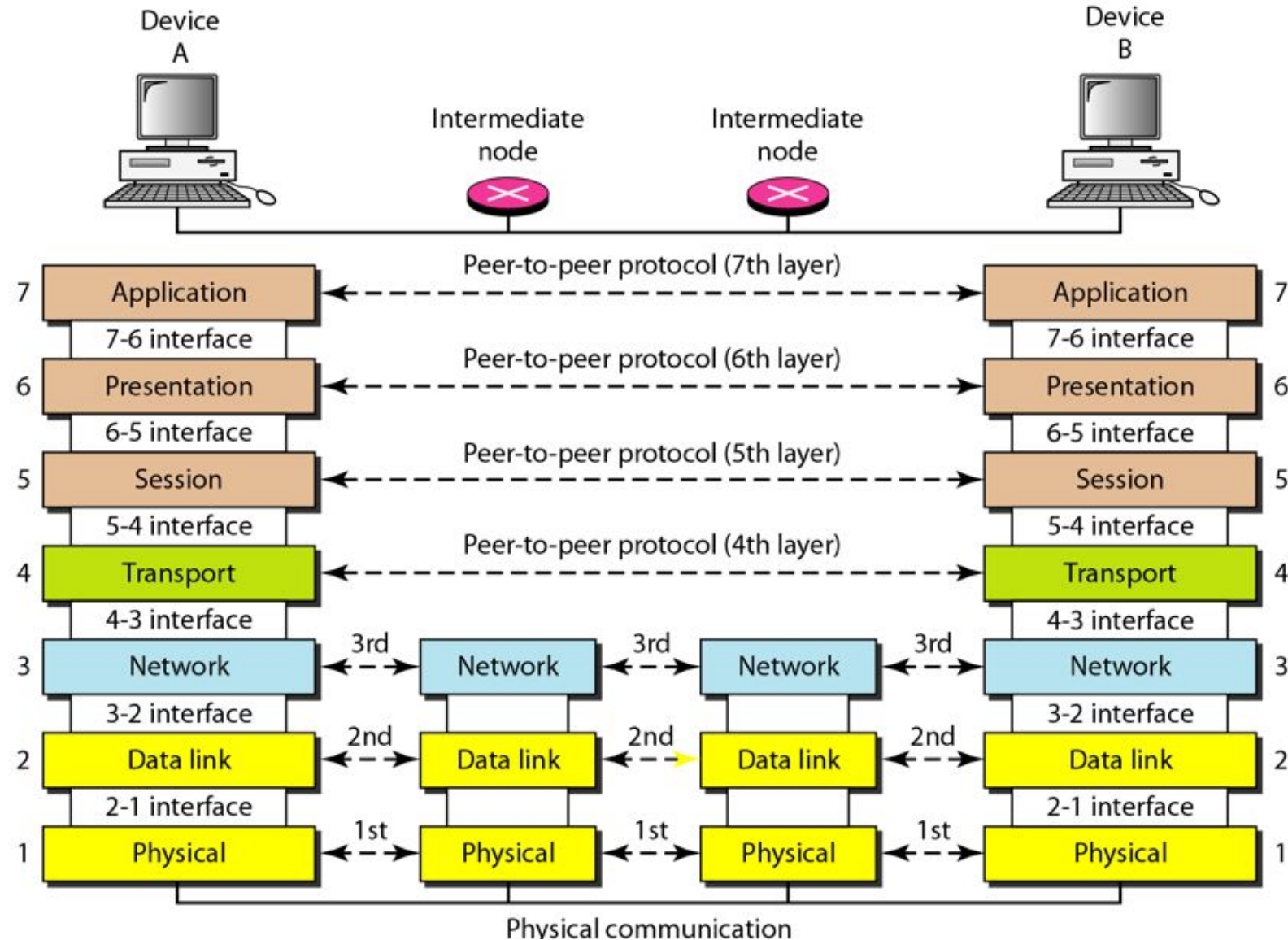
Design Issues for Layers – Flow control

- Another problem is how to keep a fast sender from bombarding a slow receiver with data
- Feedback from the receiver to the sender is often used
- This subject is called **flow control**
- Sometimes the problem is that the network is oversubscribed because too many computers want to send too much traffic, and the network cannot deliver it all
- This overloading of the network is called **congestion**

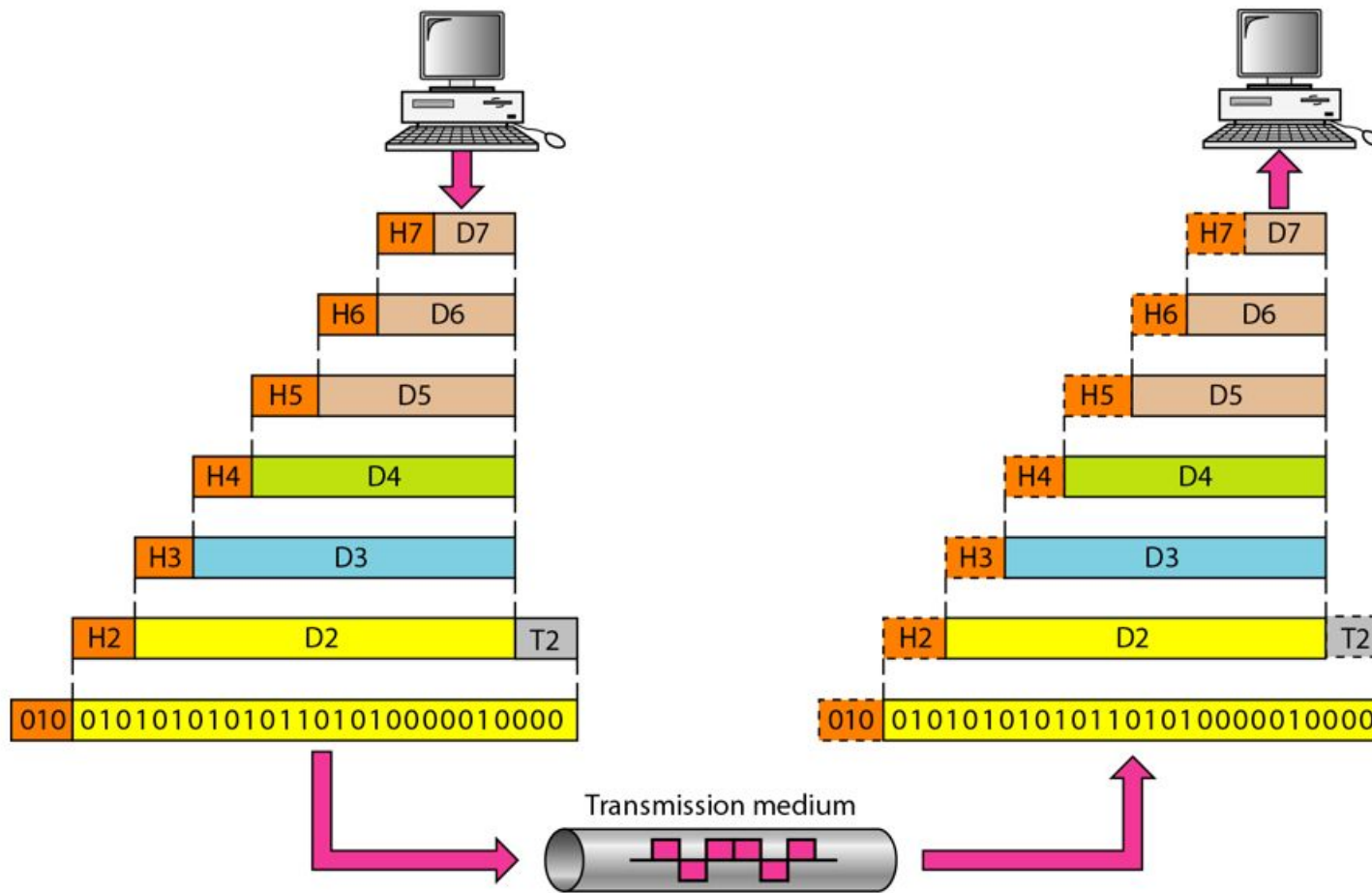
Layers in OSI



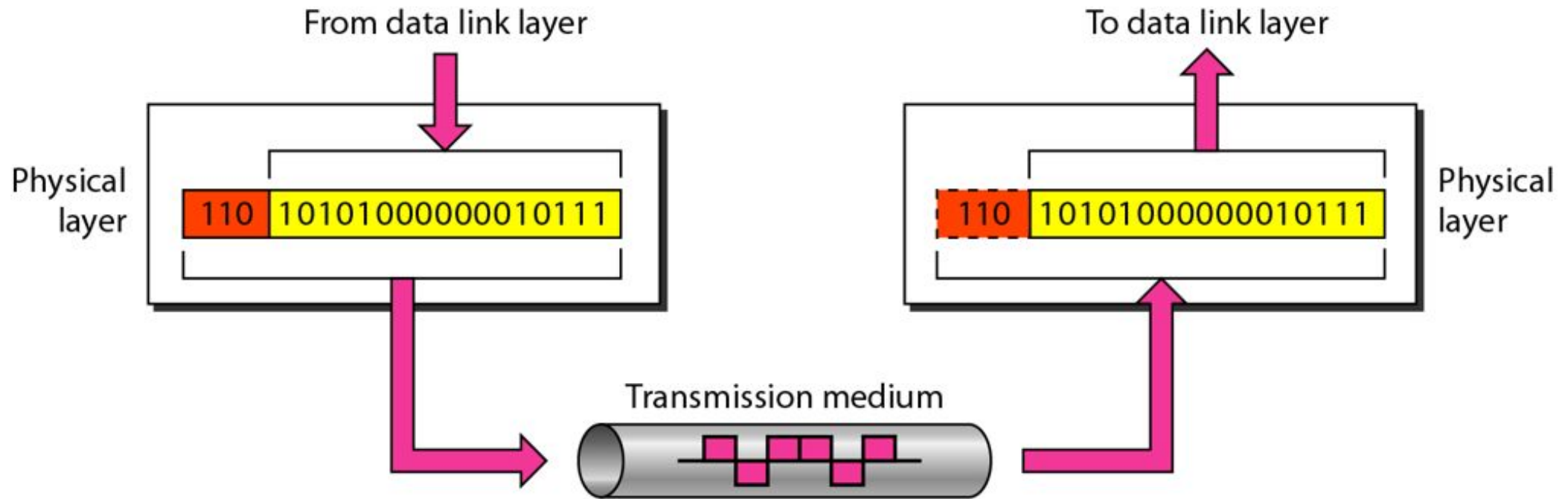
OSI Model – Interaction Between Layers



An Exchange using the OSI Model



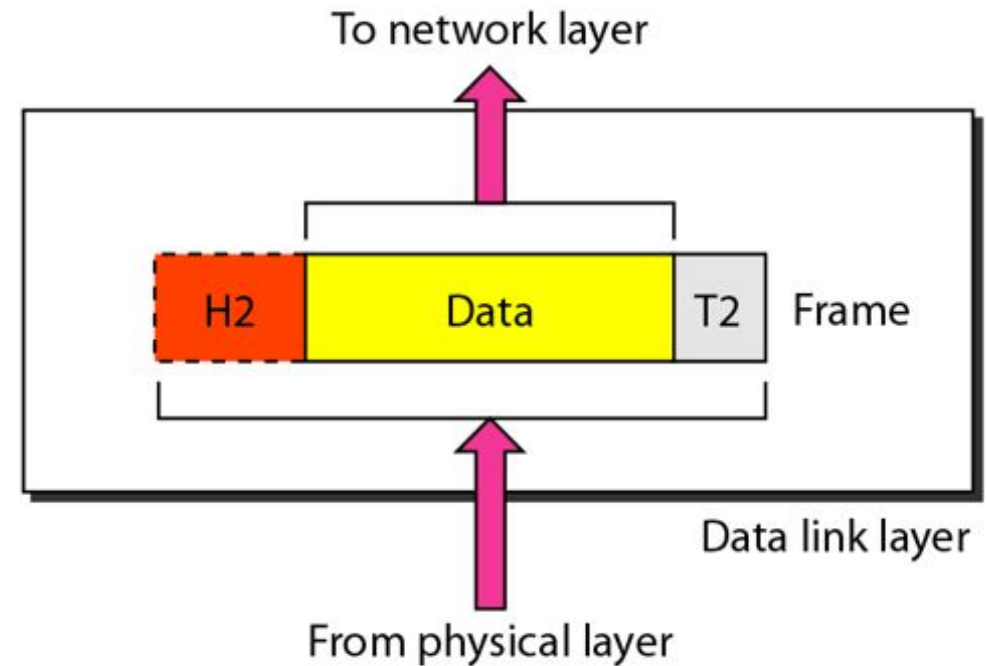
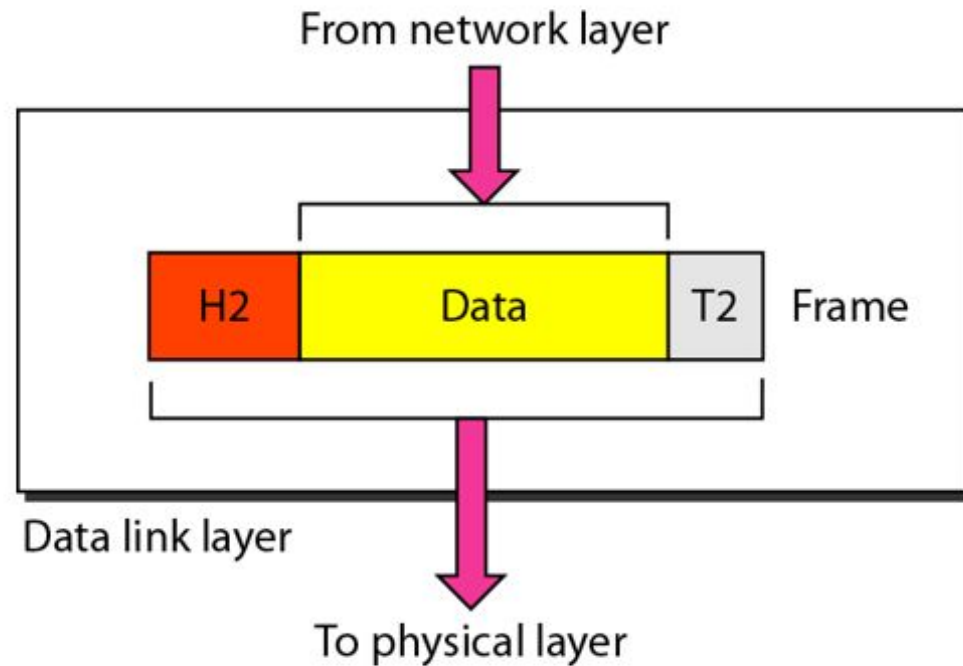
The Physical Layer



Physical Layer Functions

- Coordination of functions required to carry a bit stream over a physical medium (Analog/digital, Serial/parallel)
- Deals with mechanical and electrical specifications of the interface and transmission medium
- Encoding: Representation of **bits** into electrical or optical
- Data rate: Defines number of bits to be sent in a second (i.e. **transmission rate**)
- Synchronization: Synchronization of the clocks of the sender and the receiver (Synchronous/Asynchronous)
- Line configuration: Connection of devices to media (point-to-point or multi-point)
- Physical topology: Mesh, star, ring, ...
- Transmission mode: Simplex, half-duplex, full-duplex

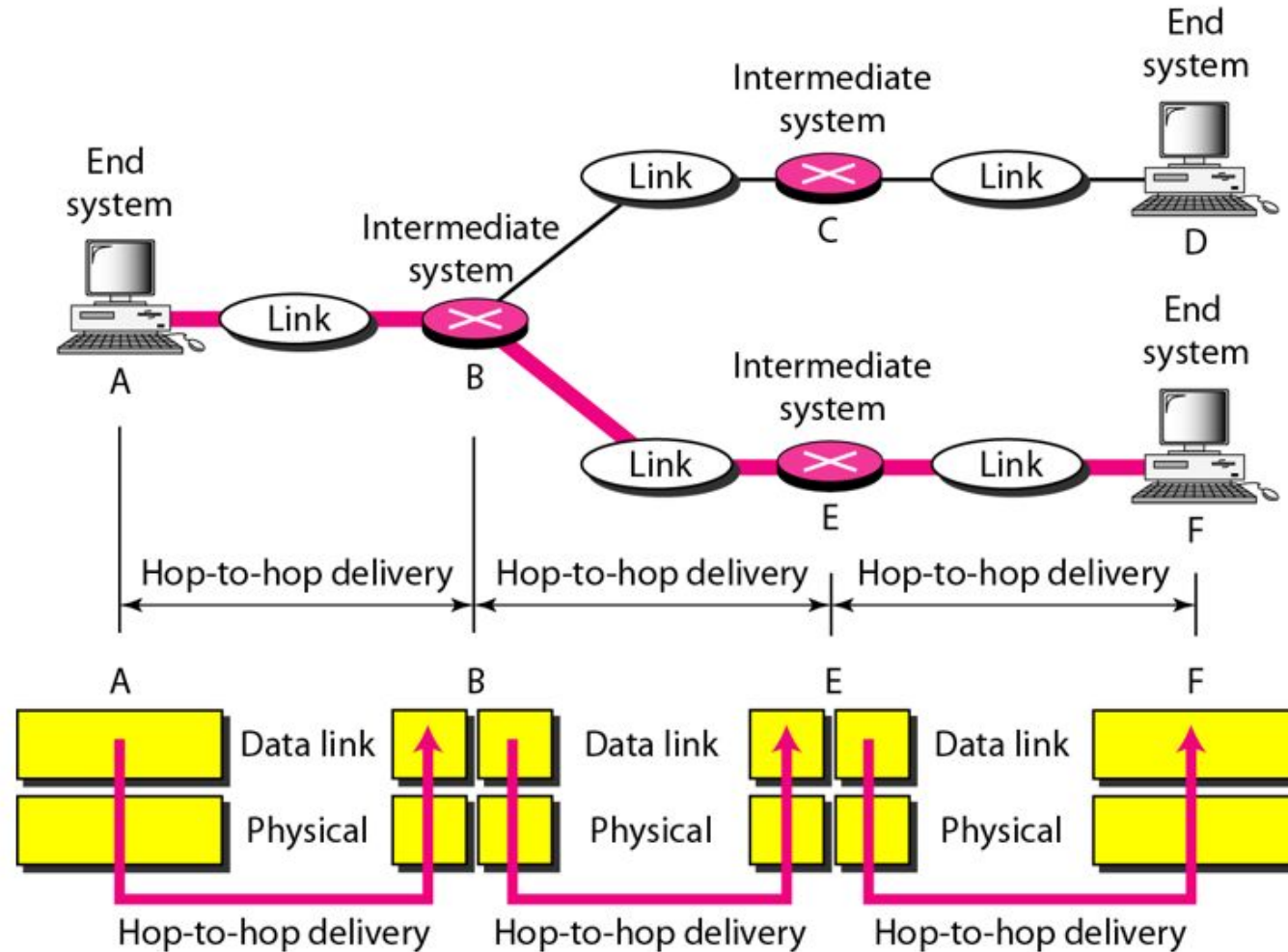
The Data Link Layer



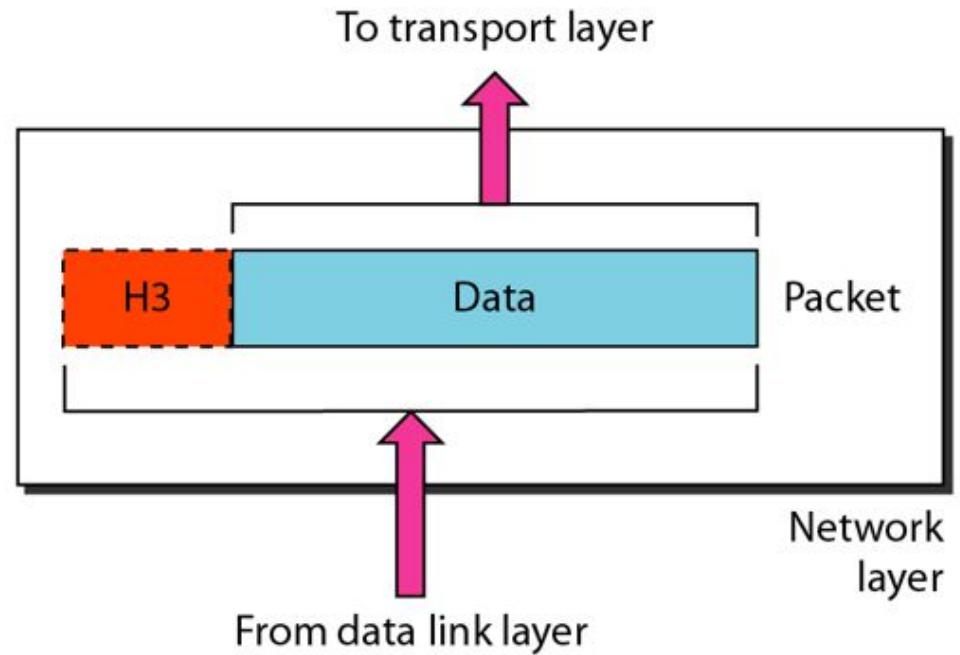
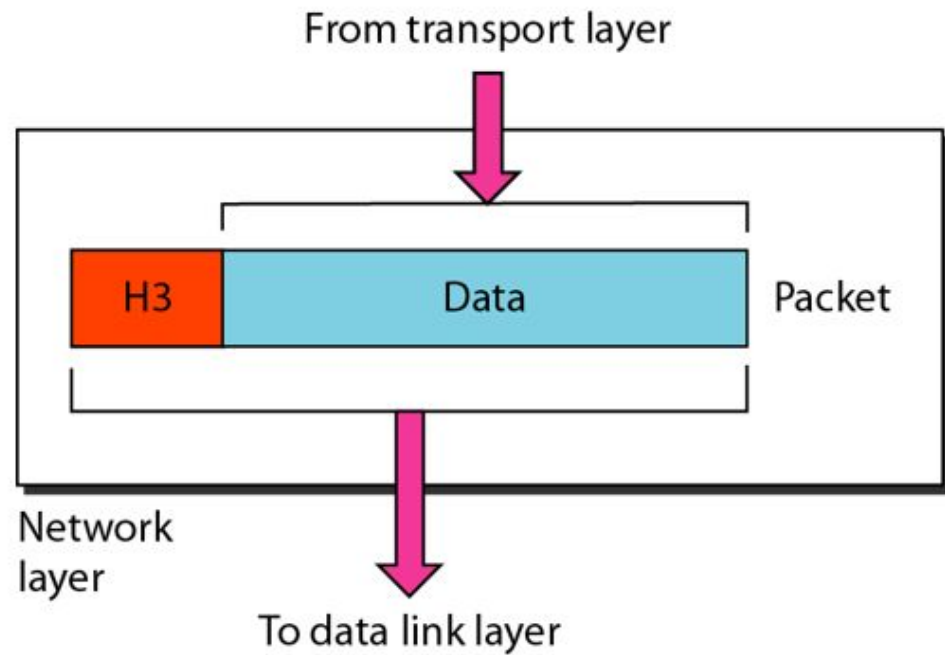
Data Link Layer

- Responsible for delivery of packets between two hosts connected to the same physical network (**hop-to-hop** – see next slide)
 - Cannot work across networks
- Converts the raw physical layer transmission into a reliable link
- Framing: Dividing bits into frames
- Physical addressing: Adding sender's and receiver's physical addresses to the frame header
- Flow control: Decide how much data to send at one go
- Error control: Add trailer to the frame to take care of damaged, lost, or duplicate frames
- Access control: Handle simultaneous transmissions from two or more devices

Hop-to-hop Delivery



The Network Layer



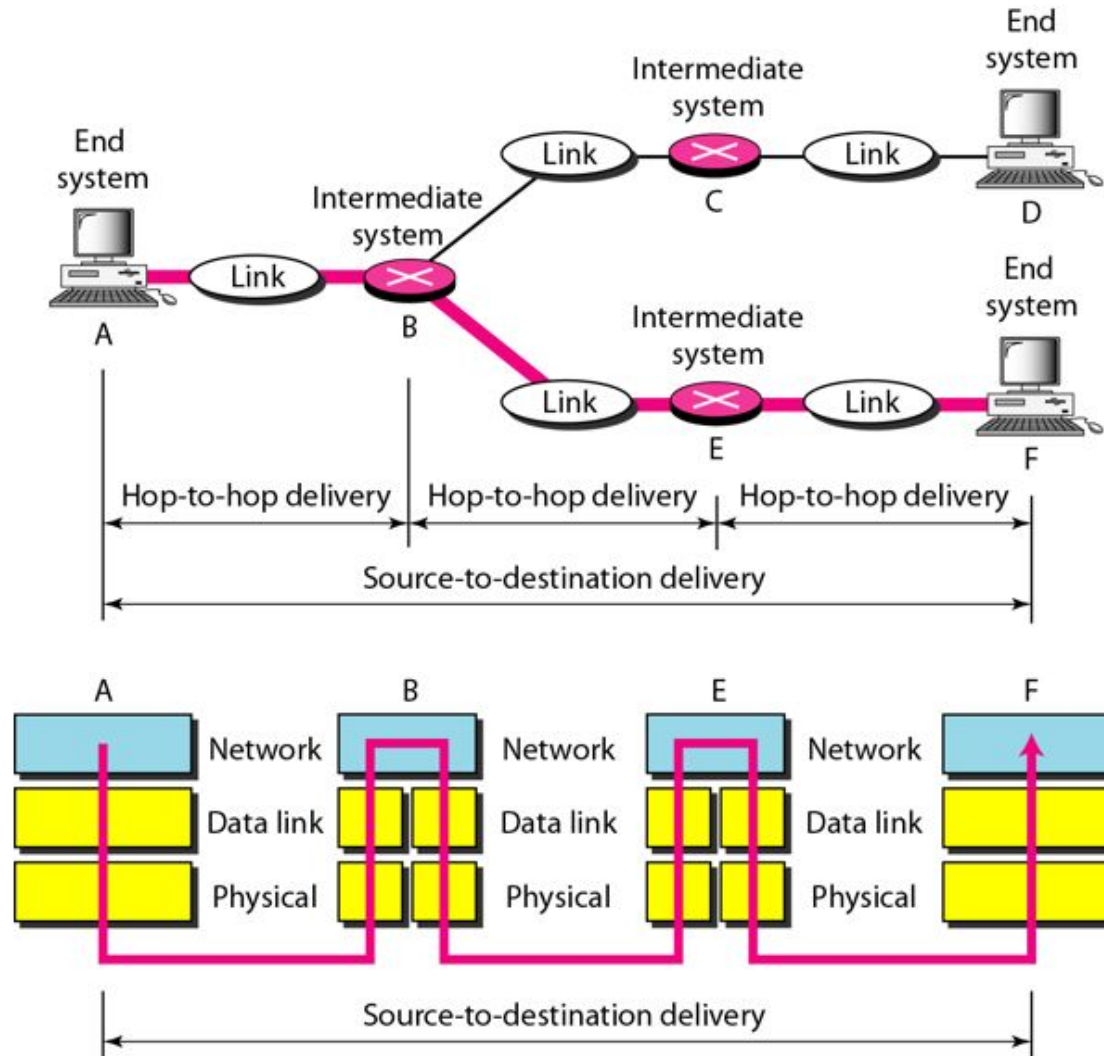
Network Layer Functions – 1

- Responsible for **source-to-destination delivery** of packets possibly across different networks (links)
 - Coordinates delivery of packets from the original sender to the final destination
 - In-between, data link layer takes care of hop-by-hop delivery between nodes attached to the same network

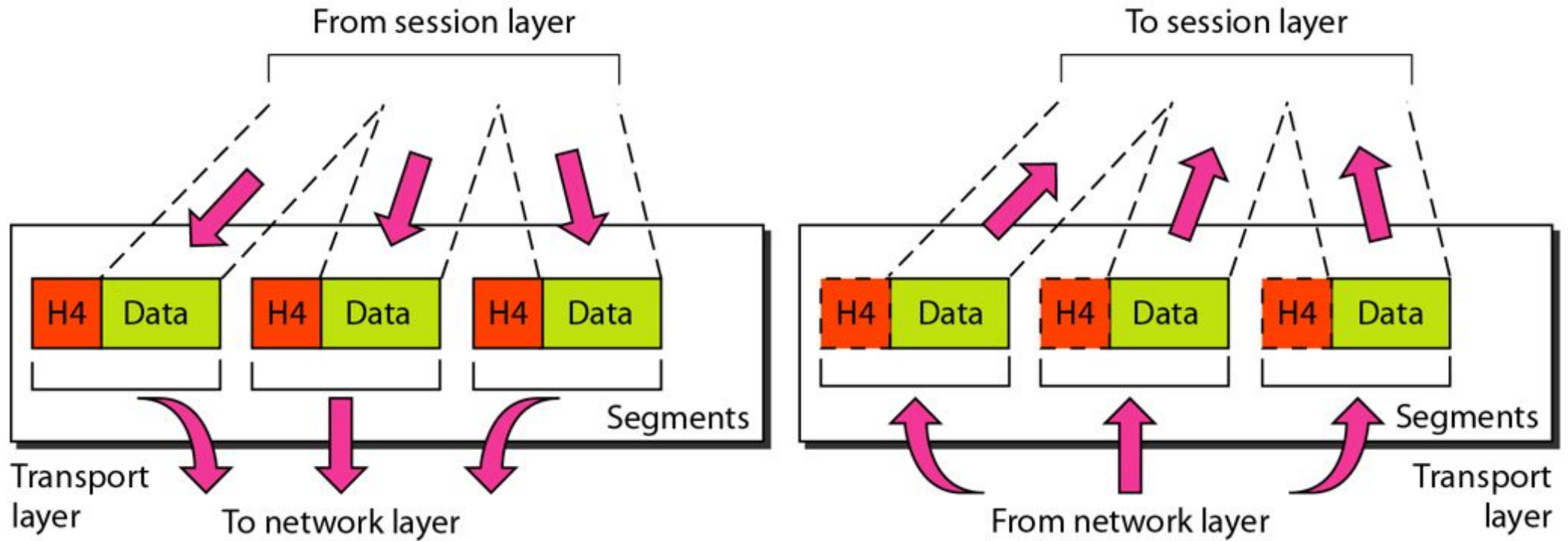
Network Layer Functions – 2

- **Logical addressing:** Physical addresses are defined by data link layer to take care of addressing inside the same network. If a packet has to travel across a network to another network, **physical addressing** will not work. Hence, a “**universal address**” or a “logical address” is required. This is done at the network layer.
 - A **header** is added for this and other purposes.
- **Routing:** When independent networks are connected to form an internetwork (or internet), i.e. a network of networks, the connecting devices (switches or routers) route the packets to the final destination. Network layer is responsible for this function.

Source-to-destination Delivery



The Transport Layer

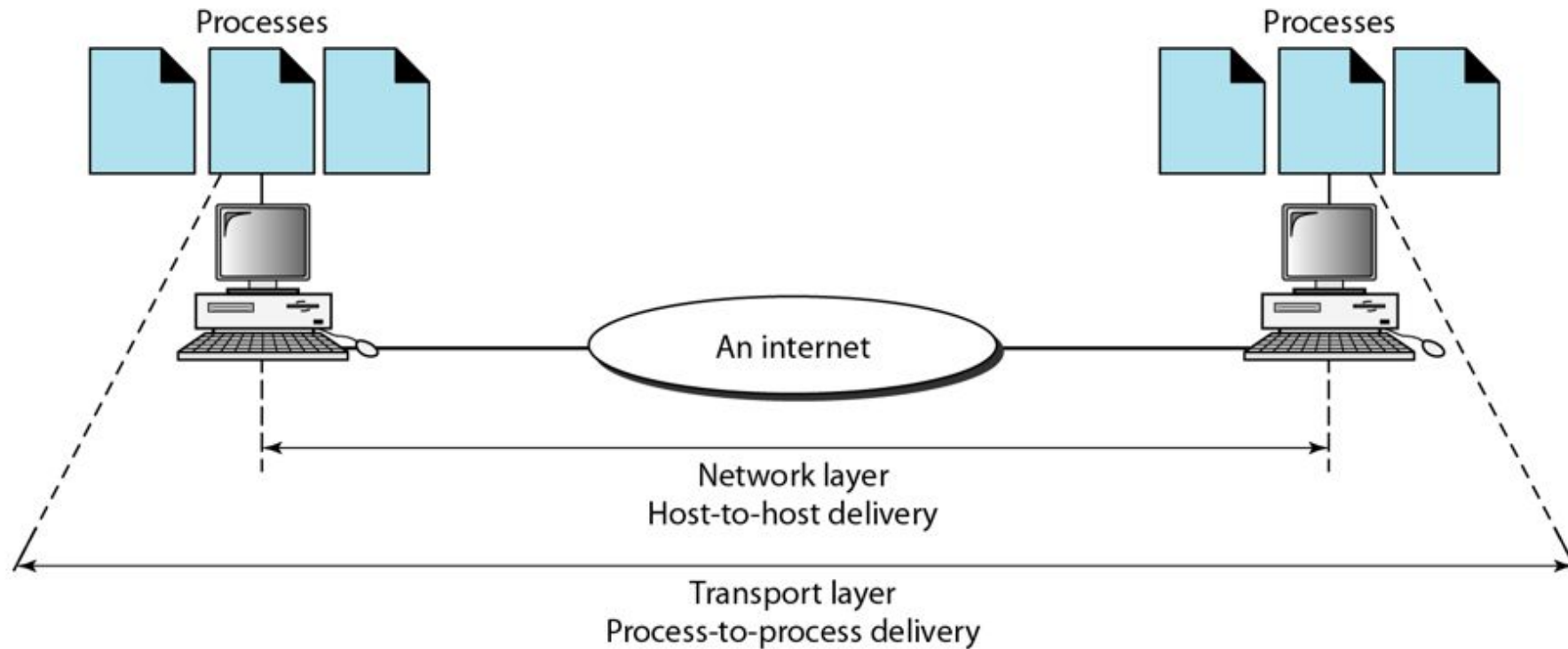


Transport Layer

- **Process** is an application program running on a host
- Network layer delivers packets from source to destination, but has no clue about the relationship between these packets
 - It treats each one of them as an independent packet
- Transport layer takes care of ensuring that the whole message arrives from the sender to the destination
 - Also handles **flow control** and **error control** at the source-to-destination level

<i>Layer</i>	<i>Responsibility</i>
<i>Data link</i>	Hop-to-hop
<i>Network</i>	Source-to-destination
<i>Transport</i>	Process-to-process

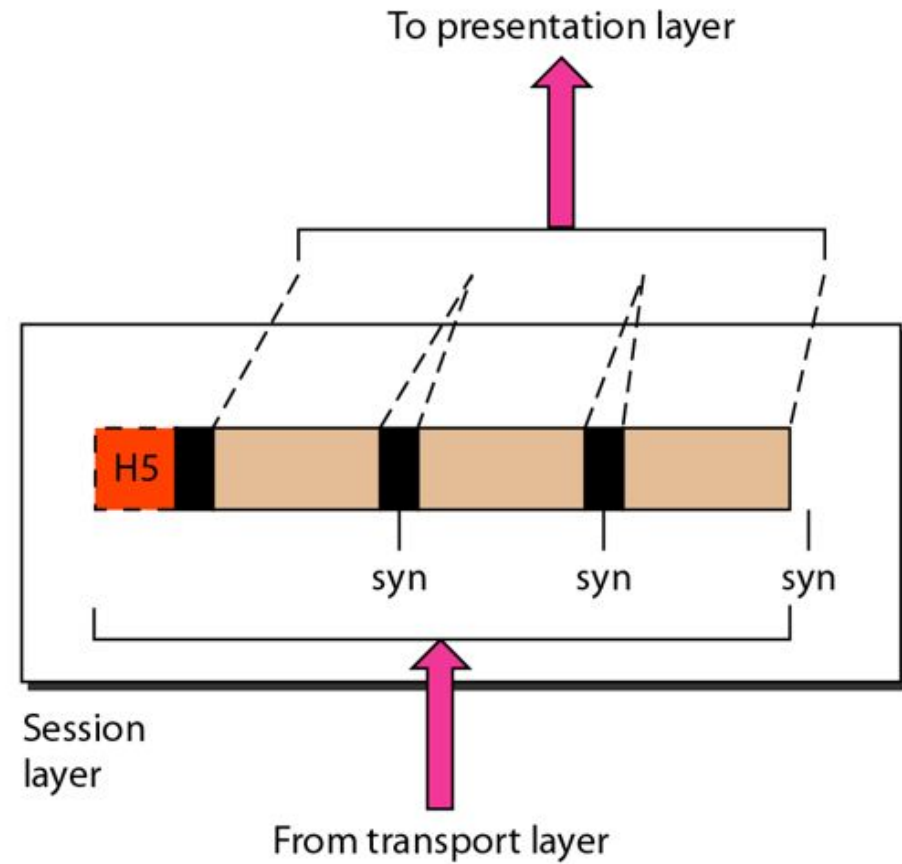
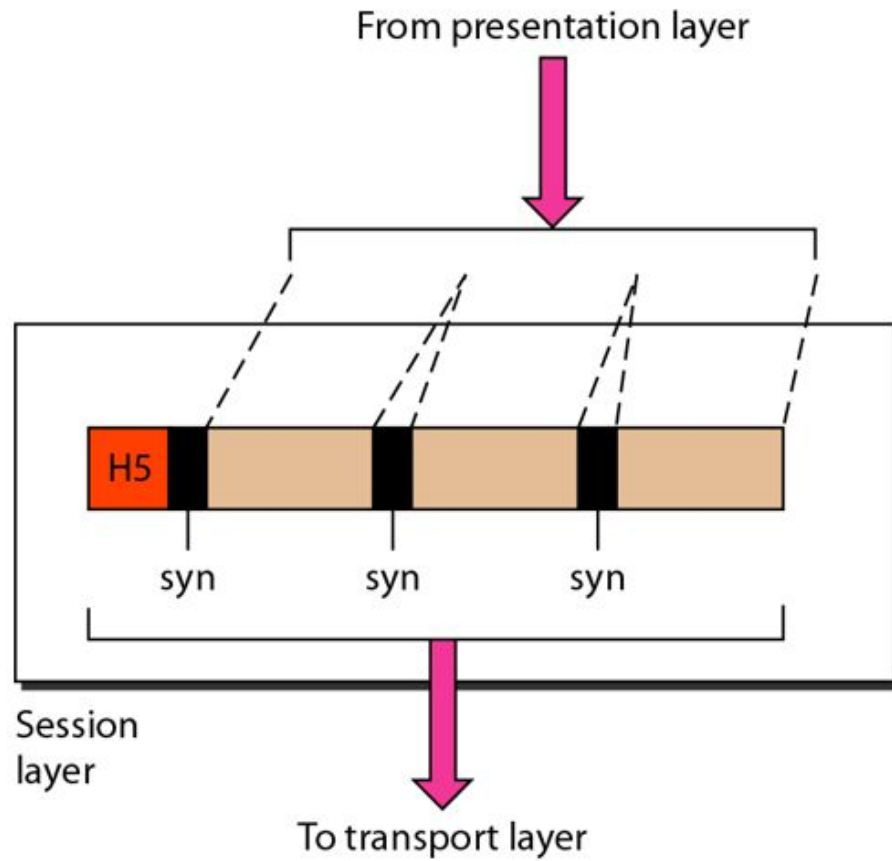
Reliable Process-to-process Delivery of a Message



Transport Layer

- **Port addressing:** Ensure the right application gets packets on a computer running several applications
- **Segmentation and reassembly:** Divide a message into segments, each bearing a segment number
 - Destination uses these to reassemble messages and to track loss, duplication, or sequencing problems
- **Connection control:** Connection-oriented or connection-less
- **Flow control:** End-to-end, not hop-to-hop
- **Error control:** Process-to-process, not hop-to-hop
 - Usually uses **acknowledgments** and **retransmissions** for this

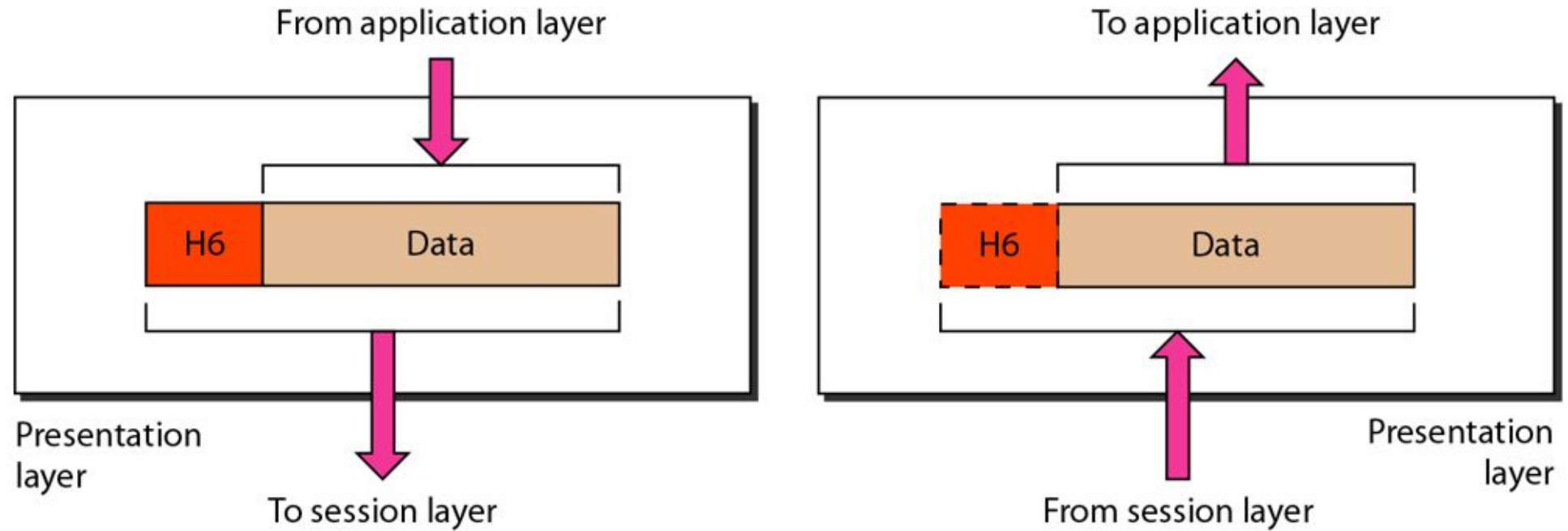
The Session Layer



Session Layer

- Dialog control: Allows two systems to enter into a dialog (half-duplex or full-duplex)
- Synchronization: Adds checkpoints to allow restart from the last checkpoint, instead of having to send the whole data once again from scratch

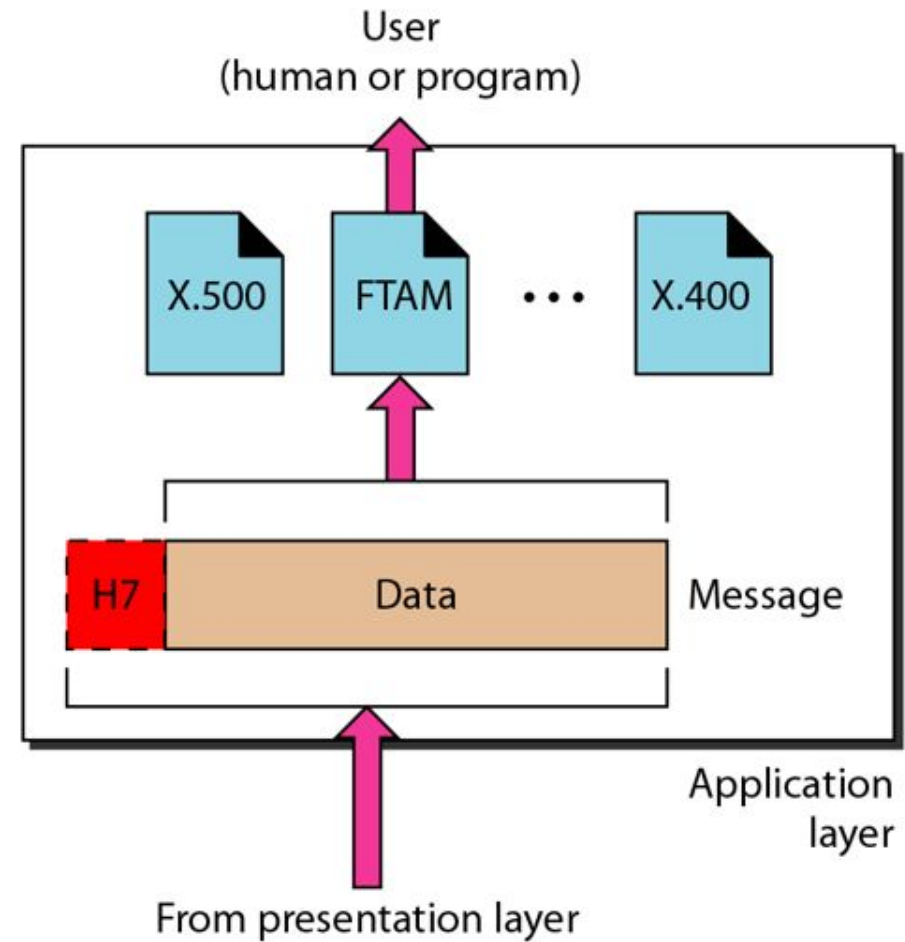
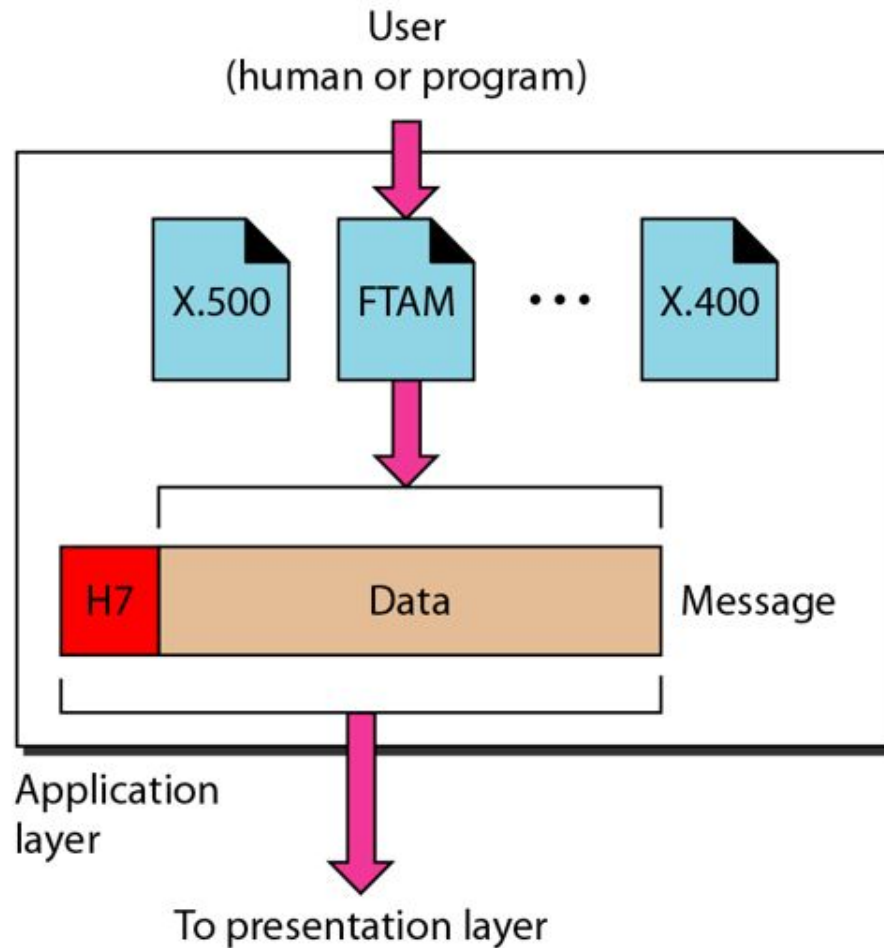
The Presentation Layer



Presentation Layer

- Translation: Use of a common format to take care of differences (**encoding**, number representation etc)
- **Encryption**: To achieve confidentiality or privacy
- **Compression**: Reduces the number of bits needed to be transmitted

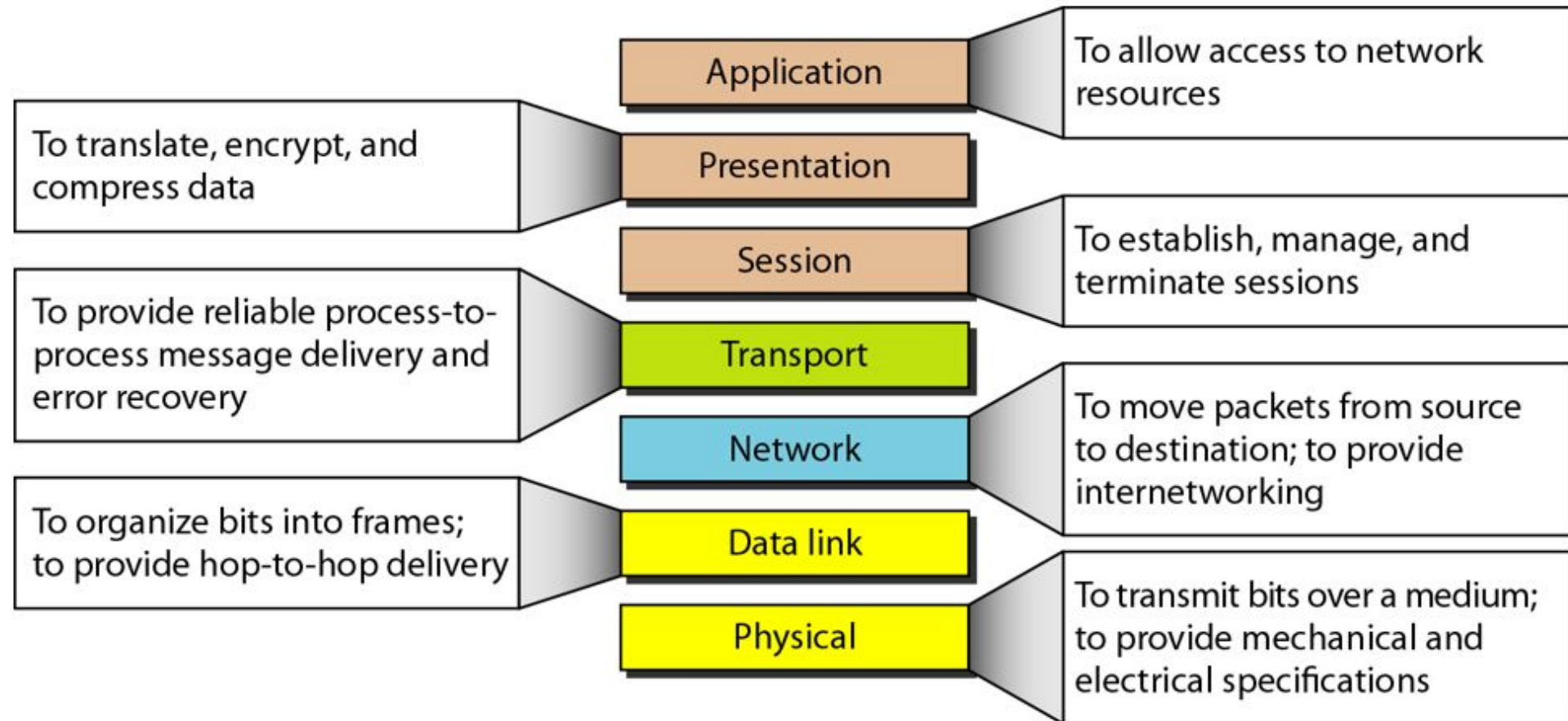
The Application Layer



Application Layer

- Provides an “application view” to a network (e.g. email, file transmission)
- Network virtual terminal: Allows a host to connect to a remote computer, as if this is a local connection
 - Creation of software emulation of a terminal at the remote computer
- File transfer, access, and management
- Mail services
- Directory services

OSI Layer Summay



TCP/IP

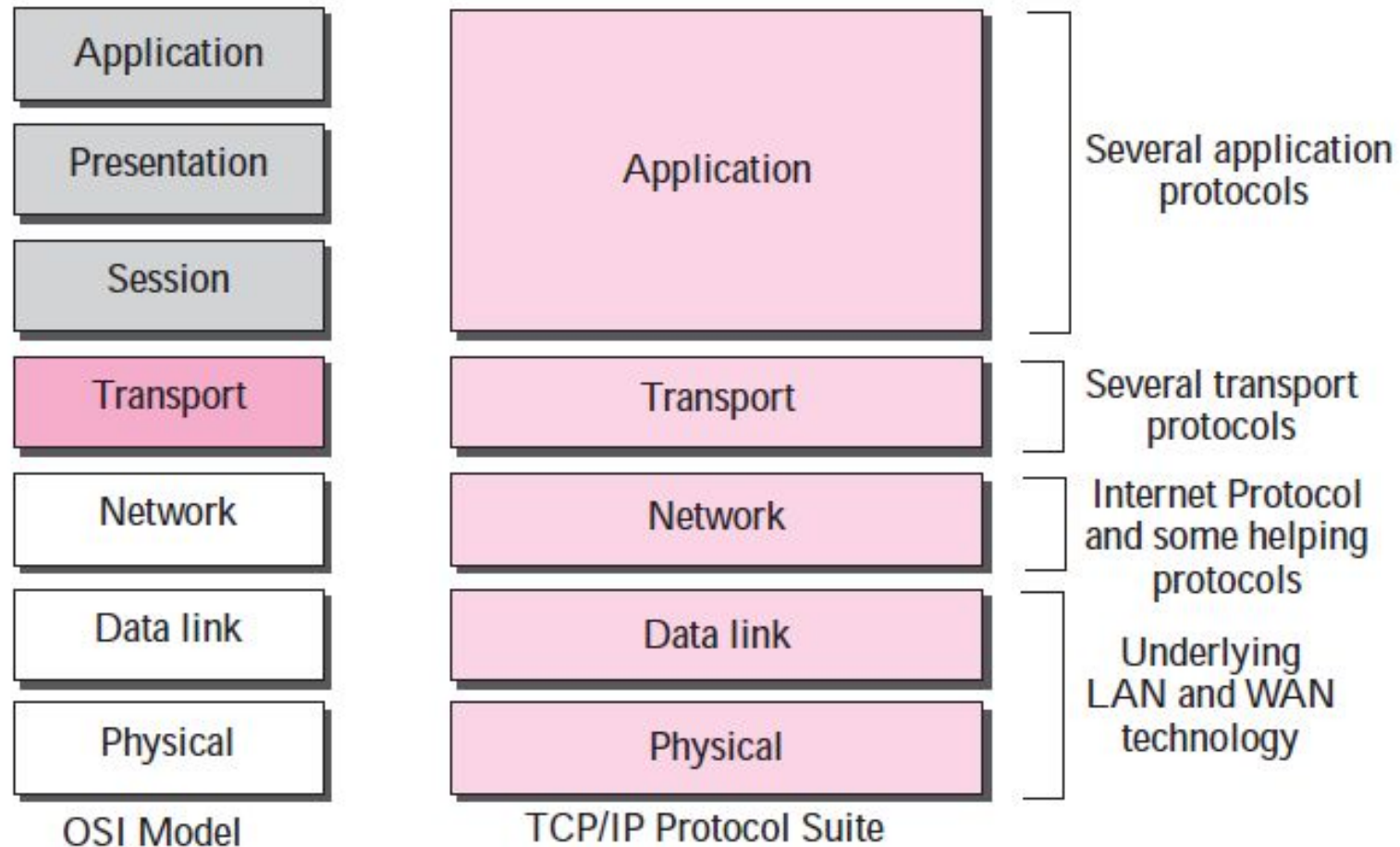
Basics of TCP/IP

- The **TCP/IP** protocol suite was developed prior to the OSI model
- Therefore, the layers in the TCP/IP protocol suite do not match exactly with those in the OSI model
- The original TCP/IP protocol suite was defined as four software layers built upon the hardware
- Today, however, TCP/IP is thought of as a five-layer model with the layers named similarly to the ones in the OSI model

OSI versus TCP/IP

- When we compare the two models, we find that two layers, session and presentation, are missing from the TCP/IP protocol suite
- These two layers were not added to the TCP/IP protocol suite after the publication of the OSI model
- The application layer in the suite is usually considered to be the combination of three layers in the OSI model
- See diagram on the next slide

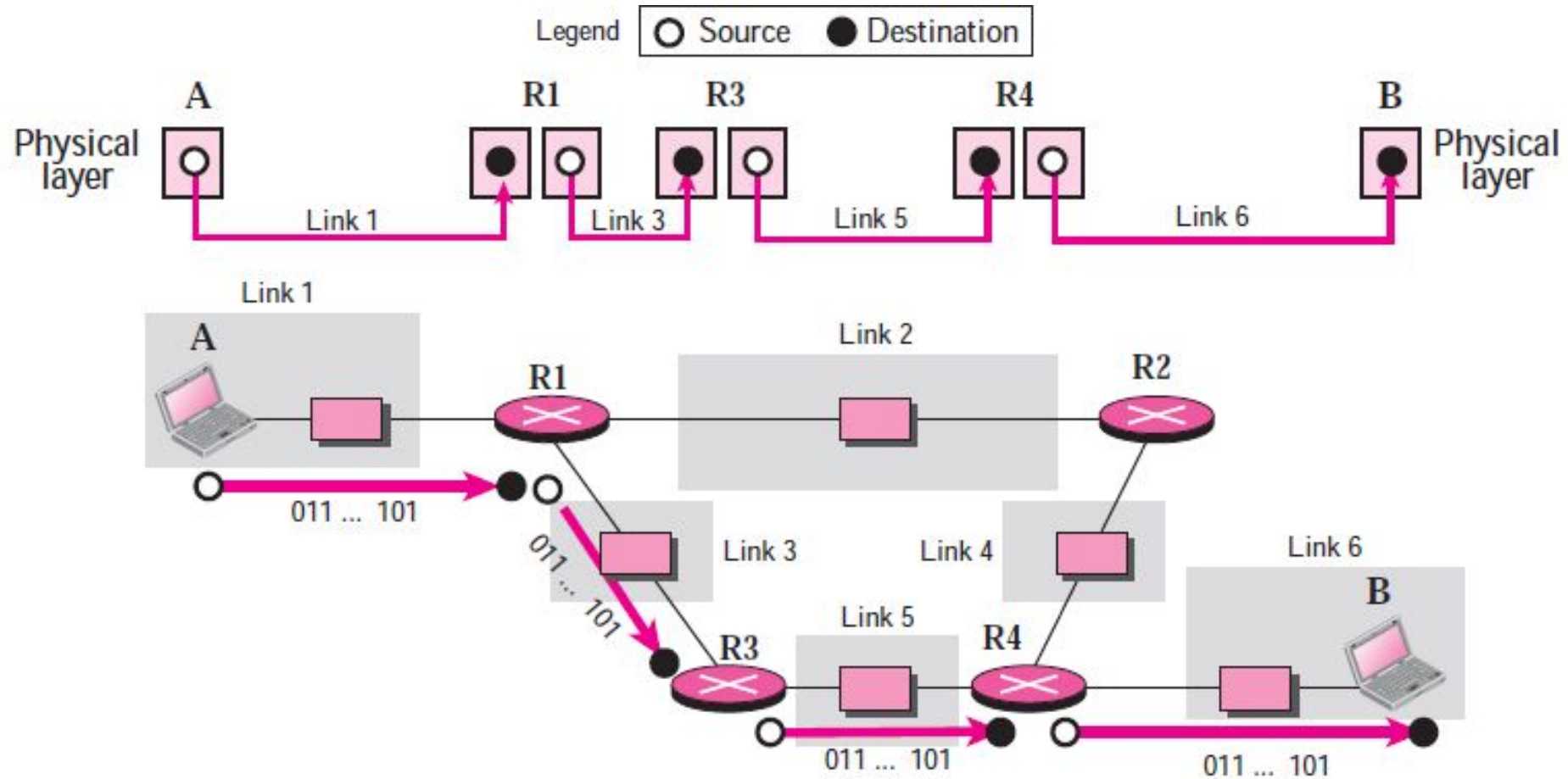
OSI versus TCP/IP



Physical Layer

- TCP/IP does not define any specific protocol for the physical layer
- It supports all of the standard and proprietary protocols
- At this level, the communication is between two hops or nodes, either a computer or router
- The unit of communication is a single bit
- When the connection is established between the two nodes, a stream of bits is flowing between them
- The physical layer, however, treats each bit individually

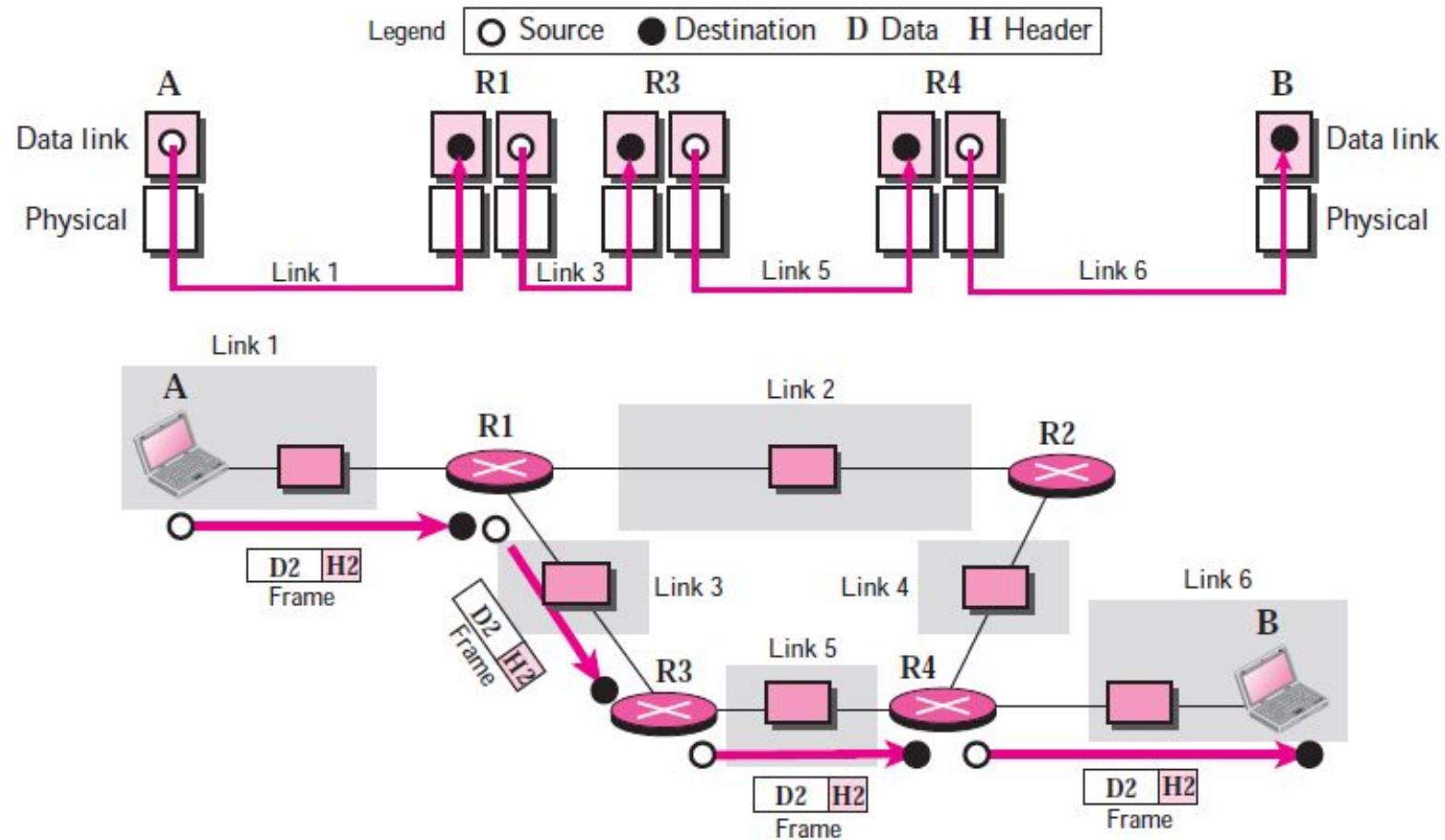
Communication at the Physical Layer



Data Link Layer

- TCP/IP does not define any specific protocol for the data link layer either
- It supports all of the standard and proprietary protocols
- At this level, the communication is also between two hops or nodes
- The unit of communication however, is a packet called a **frame**
- A frame is a packet that encapsulates the data received from the network layer with an added header and sometimes a trailer
- The header, among other communication information, includes the source and destination of frame
- The destination address is needed to define the right recipient of the frame because many nodes may have been connected to the link
- The source address is needed for possible response or acknowledgment as may be required by some protocols

Communication at the Data Link Layer



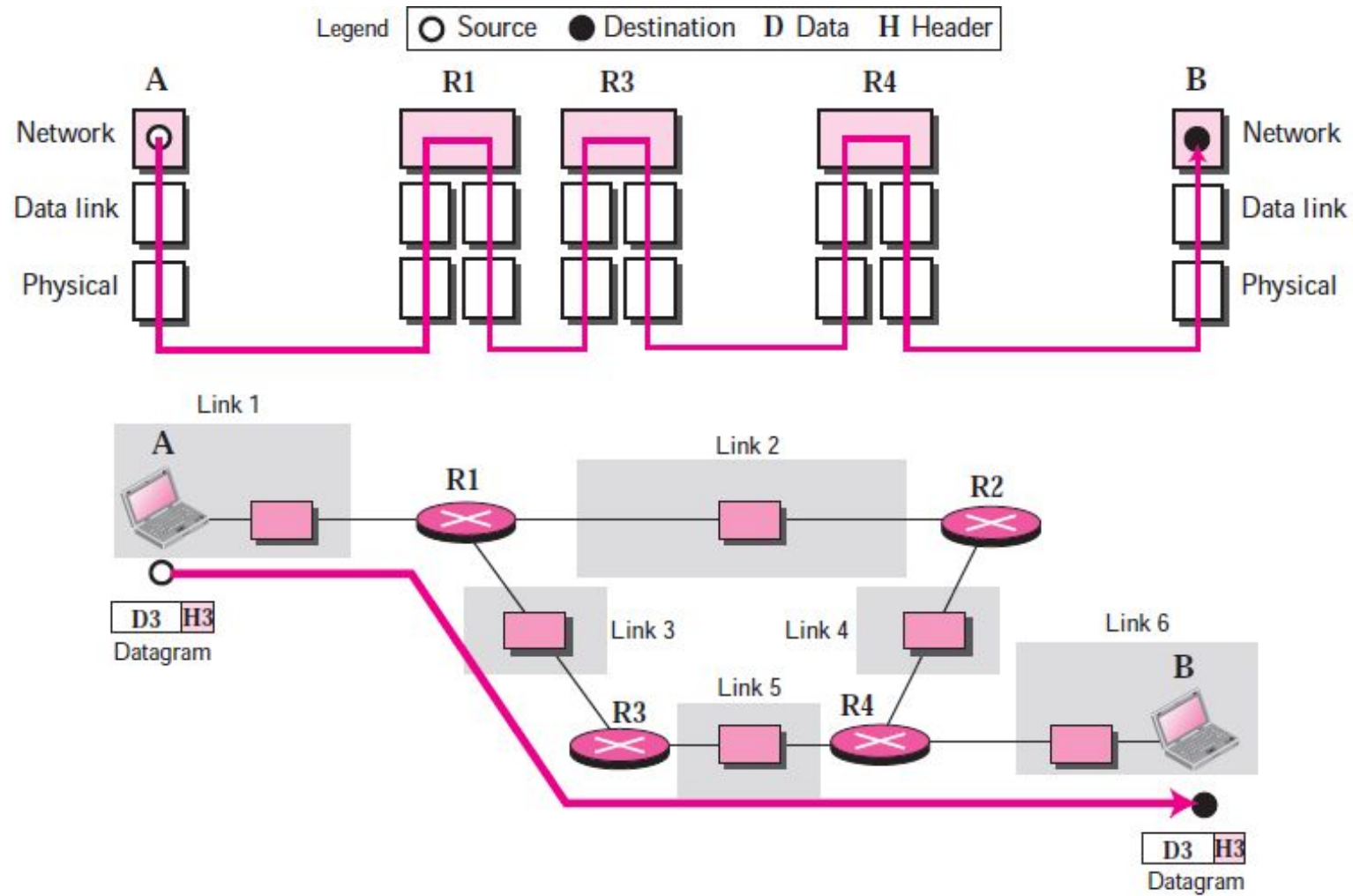
Network Layer

- At the network layer (or, more accurately, the **internetwork layer**), TCP/IP supports the **Internet Protocol (IP)**
- The Internet Protocol (IP) is the transmission mechanism used by the TCP/IP protocols
- IP transports data in packets called **datagrams**, each of which is transported separately
- Datagrams can travel along different routes and can arrive out of sequence or be duplicated
- IP does not keep track of the routes and has no facility for reordering datagrams once they arrive at their destination

Network Layer – More ...

- Note that there is a main difference between the communication at the network layer and the communication at data link or physical layers
- Communication at the network layer is end to end while the communication at the other two layers are node to node
- The datagram started at computer A is the one that reaches computer B
- The network layers of the routers can inspect the source and destination of the packet for finding the best route, but they are not allowed to change the contents of the packet
- Of course, the communication is logical, not physical
- Although the network layer of computer A and B think that they are sending and receiving datagrams, the actual communication again is done at the physical level

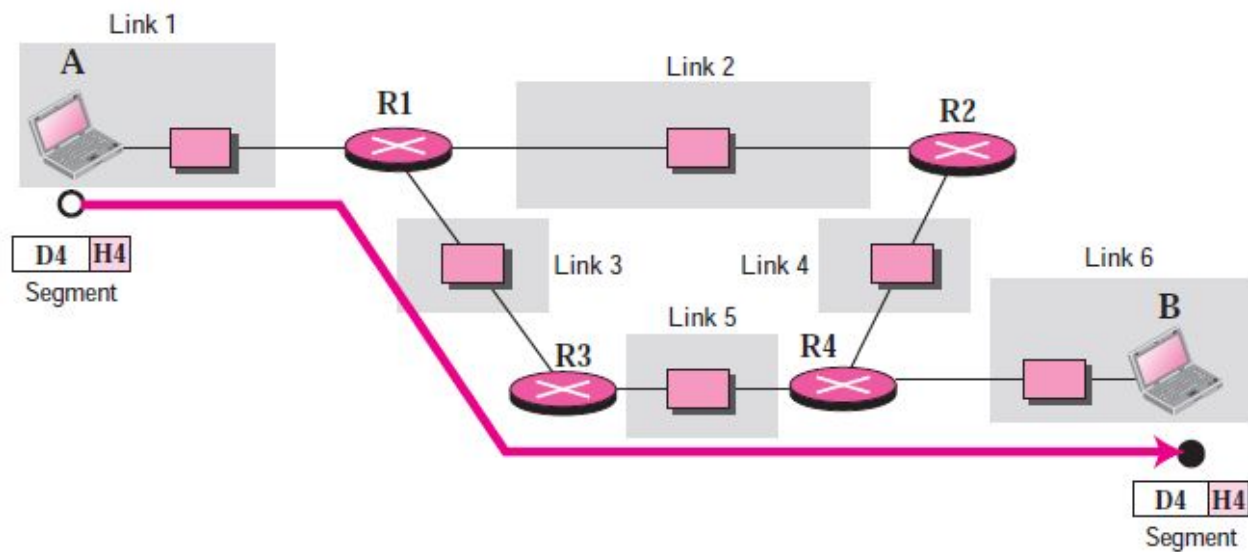
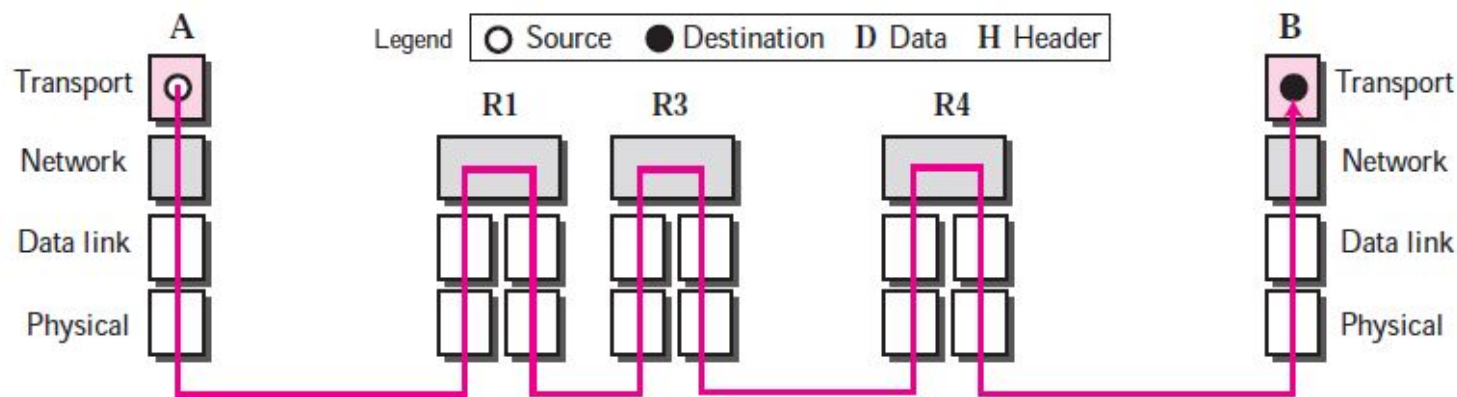
Network Layer



Transport Layer

- There is a main difference between the transport layer and the network layer
- Although all nodes in a network need to have the network layer, only the two end computers need to have the transport layer
- The network layer is responsible for sending individual datagrams from computer A to computer B; the transport layer is responsible for delivering the whole message, which is called a **segment**, a user datagram, or a packet, from A to B
- A segment may consist of a few or tens of datagrams
- The segments need to be broken into datagrams and each datagram has to be delivered to the network layer for transmission
- Since the Internet defines a different route for each datagram, the datagrams may arrive out of order and may be lost
- The transport layer at computer B needs to wait until all of these datagrams to arrive, assemble them and make a segment out of them

Transport Layer



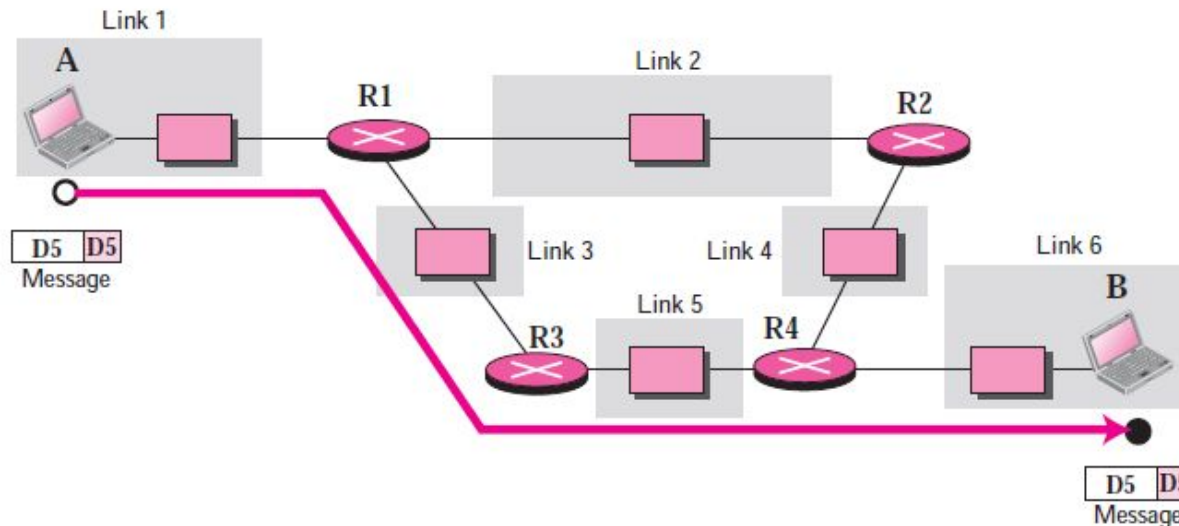
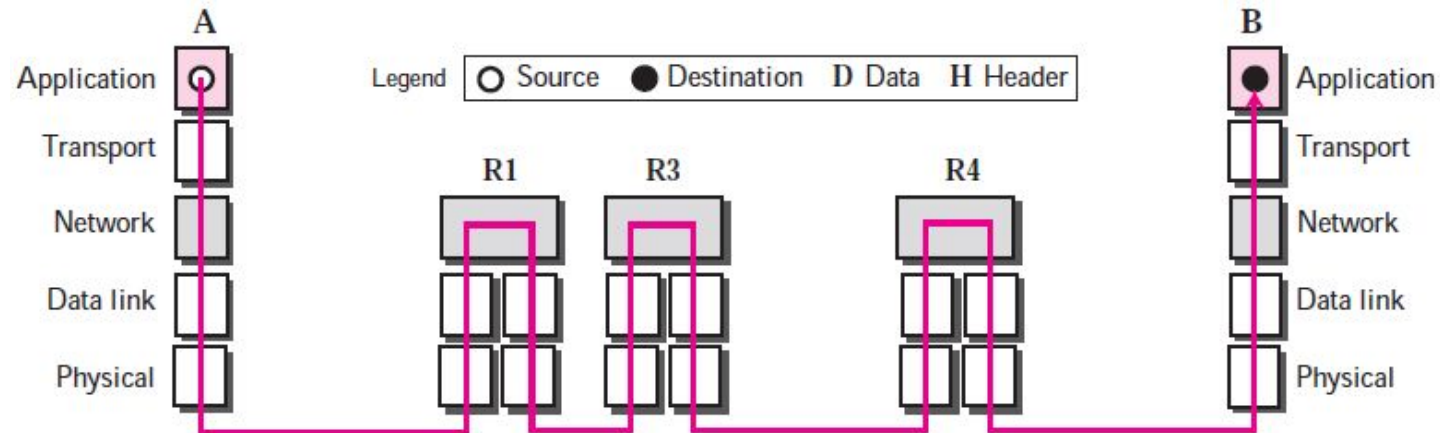
Transport Layer Protocols

- Traditionally, the transport layer was represented in the TCP/IP suite by two protocols:
 - **User Datagram Protocol (UDP)** and
 - **Transmission Control Protocol (TCP)**
- A new protocol called **Stream Control Transmission Protocol (SCTP)** has been introduced in the last few years

Application Layer

- The application layer in TCP/IP is equivalent to the combined session, presentation, and application layers in the OSI model
- The application layer allows a user to access the services of our private internet or the global Internet
- Many protocols are defined at this layer to provide services such as electronic mail, file transfer, accessing the World Wide Web, and so on

Communication at the Application Layer



Layer 2 Switching

Networking devices

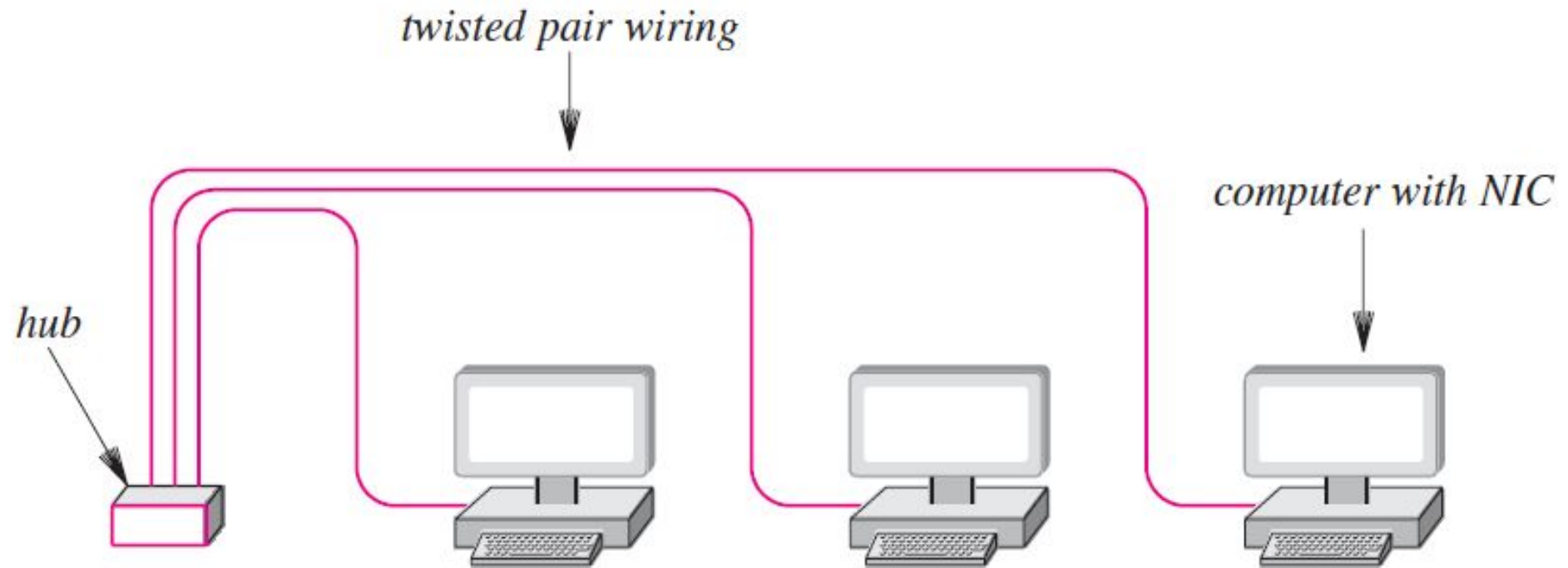
- Repeater
- Bridge
- Router
- Gateway

- Hub
- Switch

Hub

- For the original version of twisted pair Ethernet, the electronic device that served as the central interconnection was known as a **hub**.
- Hubs were available in a variety of sizes, with the cost proportional to size.
- A small hub had four or eight ports that each connected to a computer or other device (e.g., a printer).
- Larger hubs accommodated hundreds of connections.

Hub Concept



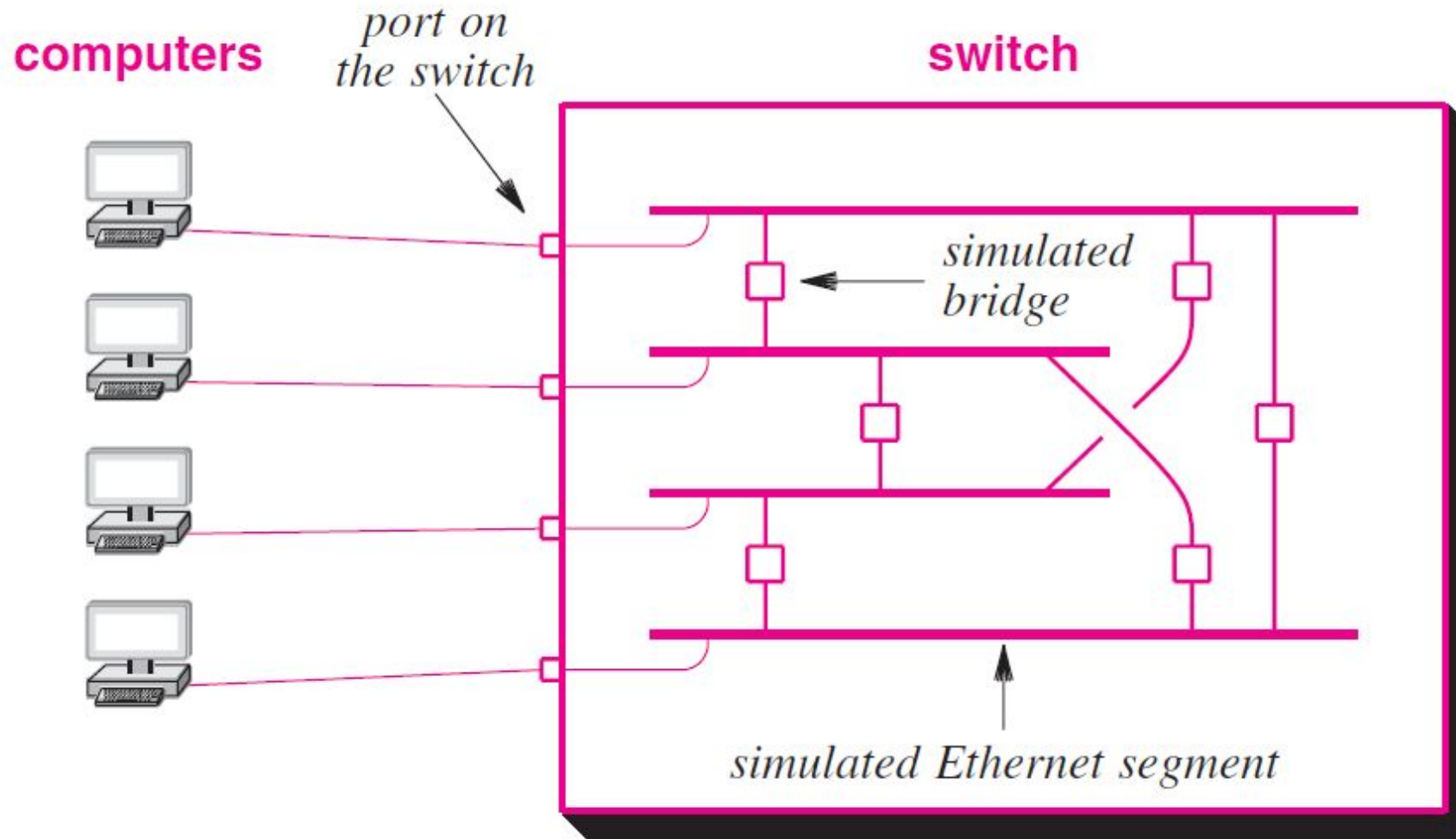
More on Hub ...

- Electronic components in a hub emulate a physical cable, making the entire system operate like a conventional Ethernet.
- For example, a computer attached to a hub uses CSMA/CD to access the network, receives a copy of each frame, and uses the address in a frame to determine whether to process or ignore the frame.

Layer 2 Switch

- An **Ethernet switch**, sometimes called a **Layer 2 switch** is an electronic device that resembles a **hub**.
- Like a hub, a switch provides multiple ports that each attach to a single computer, and a switch allows computers to send frames to one another.
- The difference between a hub and a switch arises from the way the devices operate: a hub operates as an analog device that forwards signals among computers, while a switch is a digital device that forwards packets.
- We can think of a hub as simulating a shared transmission medium, and think of a switch as simulating a bridged network that has one computer per LAN segment. (See figure on the next slide)

Switched LAN



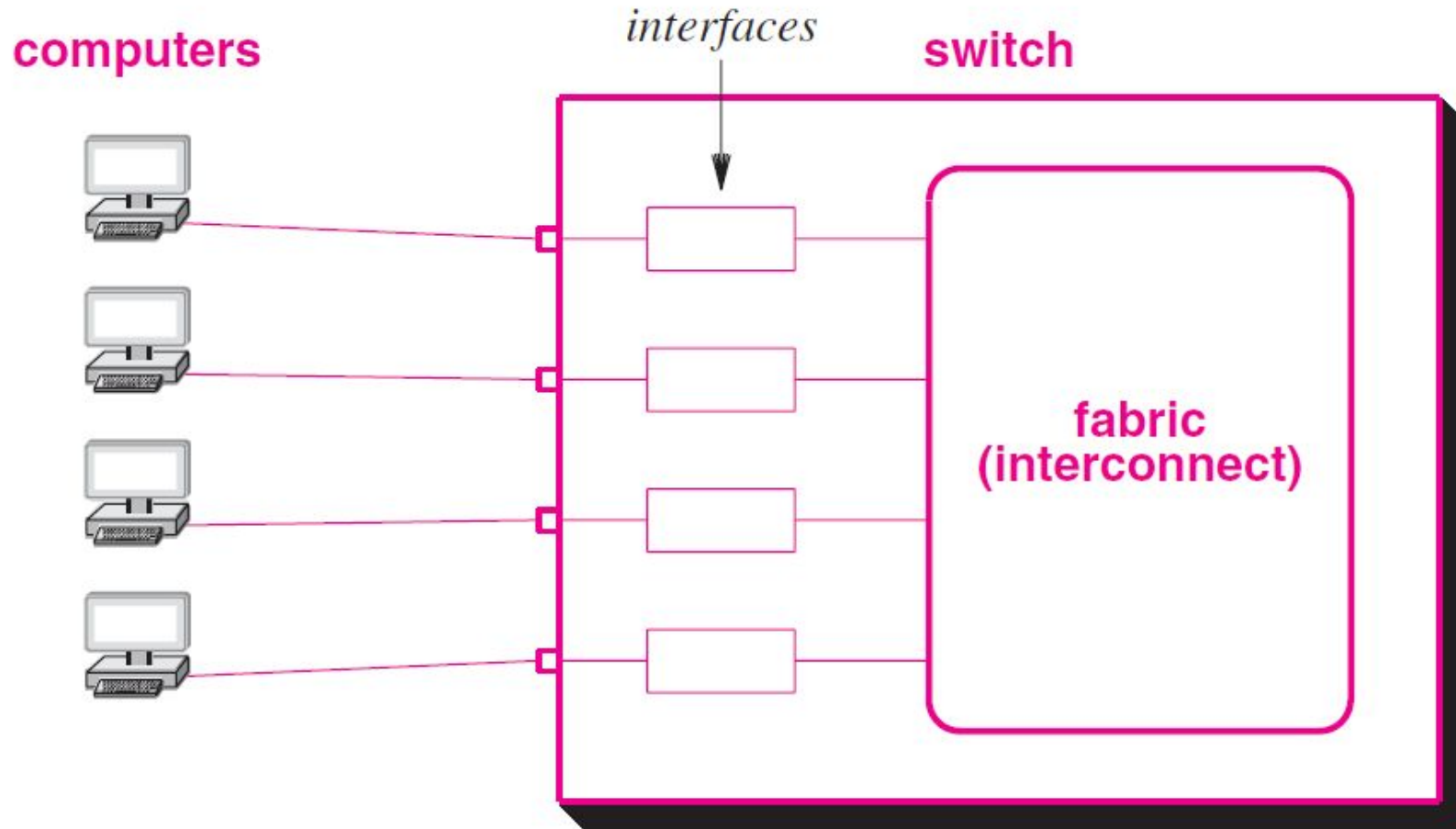
Hub versus Switch

- The main advantage of using a switched LAN instead of a hub is parallelism.
- Although a hub can only support one transmission at a time, a switch permits multiple transfers to occur at the same time, provided the transfers are independent (i.e., only one packet is being transferred to a given port at a given time).
- Thus, if a switch has N ports connected to N computers, $N/2$ transfers can occur at the same time.

More on Switch

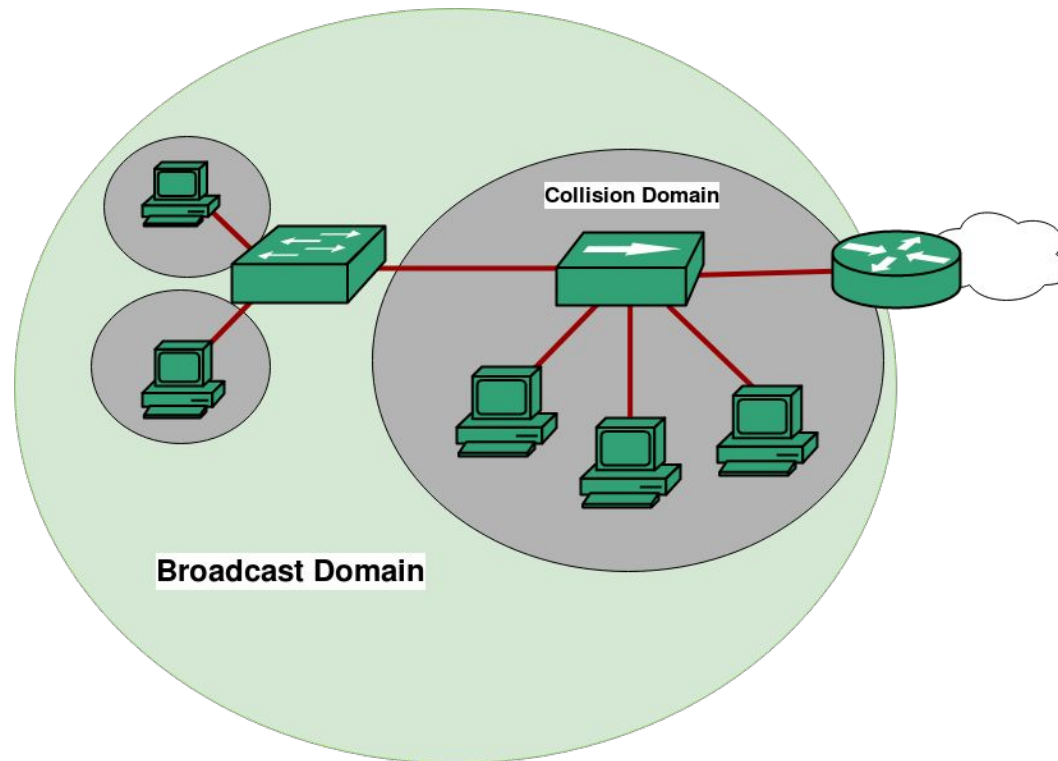
- A switch does not contain separate bridges.
- Instead, a switch consists of an intelligent interface attached to each port and a central fabric that provides simultaneous transfer between pairs of interfaces.
- An interface contains a processor, memory, and other hardware needed to accept an incoming packet, consult a forwarding table, and send the packet across the fabric to the correct output port. The interface also accepts packets from the fabric, and transmits them out the port.
- Most important, because it contains memory, an interface can buffer arriving packets when an output port is busy. Thus, if computer 1 and computer 2 send packets to computer 3 simultaneously, either interface 1 or interface 2 will hold a packet while the other interface transmits. (See figure)

Switch Architecture



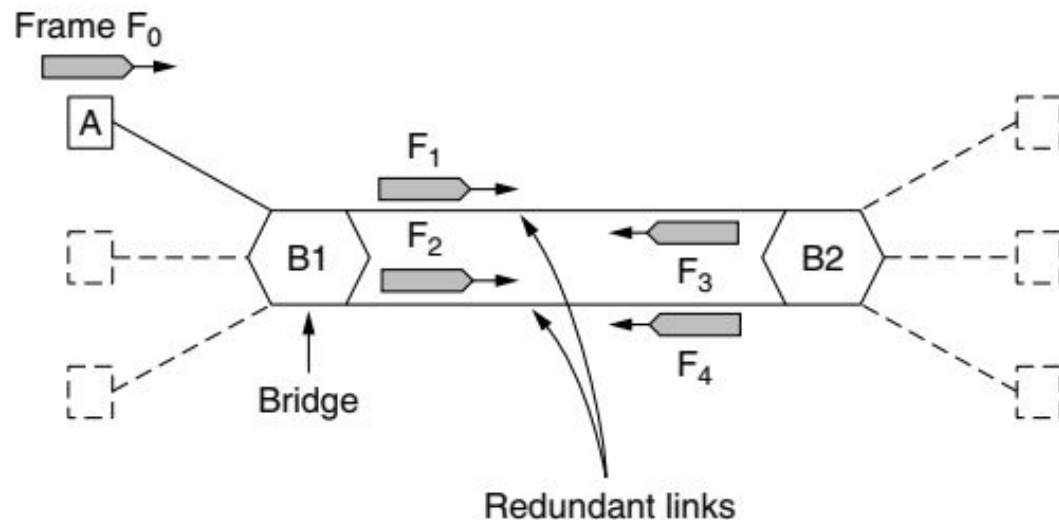
Collision domain concept

- All devices that use the same physical media and listen to each frame are considered to be in the same **collision domain**.
- This means that only one device can transmit at any given time, and any other device on the network segment must synchronize with the signal and extract the frame.



Spanning Tree Protocol (STP)

- To increase reliability, redundant links can be used between bridges.
- Let us consider two parallel links between a pair of bridges.
- This design ensures that if one link is cut, the network will not be partitioned into two sets of computers that cannot talk to each other.



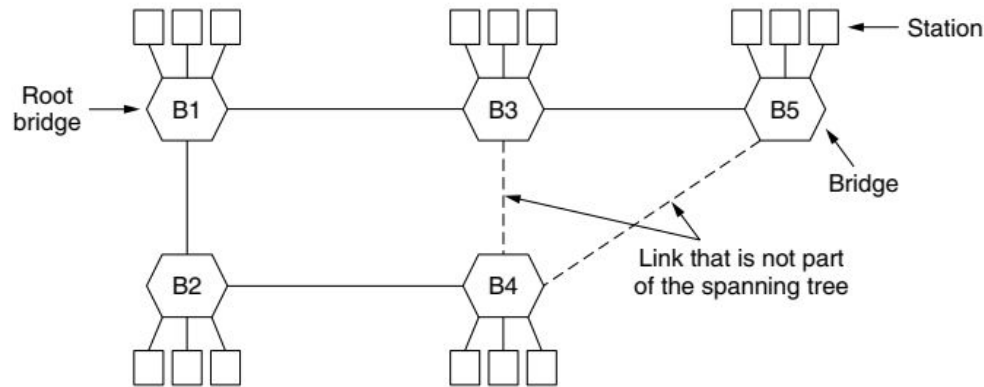
Spanning Tree Protocol (STP) – Problem

- This design solves one problem, but can create another – loops.
- Suppose host A sends a frame F0 that reaches bridge B1. The bridge will send copies of this frame to all its other ports, using the usual flooding concept.
- Since there are two links between B1 and B2, two copies of the frame, called F1 and F2 will reach B2.
- B2 does not know F1 and F2 are copies of each other, so it copies them as F3 and F4 and floods, meaning they will also go back to B1.
- B1 again sees two new frames with an unknown destination and repeats the earlier steps; B2 does the same – this will go on and on

Spanning Tree Protocol (STP) – Solution

- The solution to this difficulty is for the bridges to communicate with each other and overlay the actual topology with a **spanning tree** that reaches every bridge.
- In effect, some potential connections between bridges are ignored in for constructing a virtual loop-free topology that is a subset of the actual topology.
- Let us consider a spanning tree containing 5 bridges.

Spanning Tree Example



- There are some redundant connections between the bridges so that frames will be forwarded in loops if all of the links are used.
- This topology can be thought of as a graph in which the bridges are the nodes and the point-to-point links are the edges.
- The graph can be reduced to a spanning tree, which has no cycles by dropping the links shown as dashed lines.
- Using this spanning tree, there is exactly one path from every station to every other station.
- Once the bridges have agreed on the spanning tree, all forwarding between stations follows the spanning tree.
- Since there is a unique path from each source to each destination, loops are impossible.

Construction of the spanning tree – 1

- The bridges must first choose one bridge to be the root of the spanning tree.
- To make this choice, they each include an identifier based on their MAC address in the configuration message, as well as the identifier of the bridge they believe to be the root.
- MAC addresses are installed by the manufacturer and guaranteed to be unique worldwide, which makes these identifiers convenient and unique. The bridges choose the bridge with the lowest identifier to be the root. After enough messages have been exchanged to spread the news, all bridges will agree on which bridge is the root.
- In our example, bridge B1 becomes the root.

Construction of the spanning tree – 2

- Next, a tree of shortest paths from the root to every bridge is constructed. Here, bridges B2 and B3 can be reached from bridge B1 in a single hop, that is a shortest path.
- Bridge B4 can be reached in two hops, via either B2 or B3. To break this tie, the path via the bridge with the lowest identifier is chosen, so B4 is reached via B2.
- Bridge B5 can be reached in two hops via B3.
- To find these shortest paths, bridges include the distance from the root in their configuration messages. Each bridge remembers the shortest path it finds to the root. The bridges then turn off ports that are not part of the shortest path.
- Although the tree spans all the bridges, not all the links (or even bridges) are necessarily present in the tree. This happens because turning off the ports prunes some links from the network to prevent loops. Even after the spanning tree has been established, the algorithm continues to run during normal operation to automatically detect topology changes and update the tree.

IPv6

IPv6 addressing

- An IPv6 address is 128 bits (16 bytes) long, which is 4 times that of IPv4
- Instead of the **dotted decimal notation**, in IPv6, we use **colon hexadecimal notation**

221.14.65.11.105.45.170.34.12.234.18.0.14.0.115.255

FDEC : BA98 : 7654 : 3210 : ADBF : BBFF : 2922 : FFFF

- Here, we have 8 sections of 2 bytes each, each 2 bytes represented as 4 hex digits – hence we have 32 hex digits for an IPv6 address
- **Zero compression** can be used



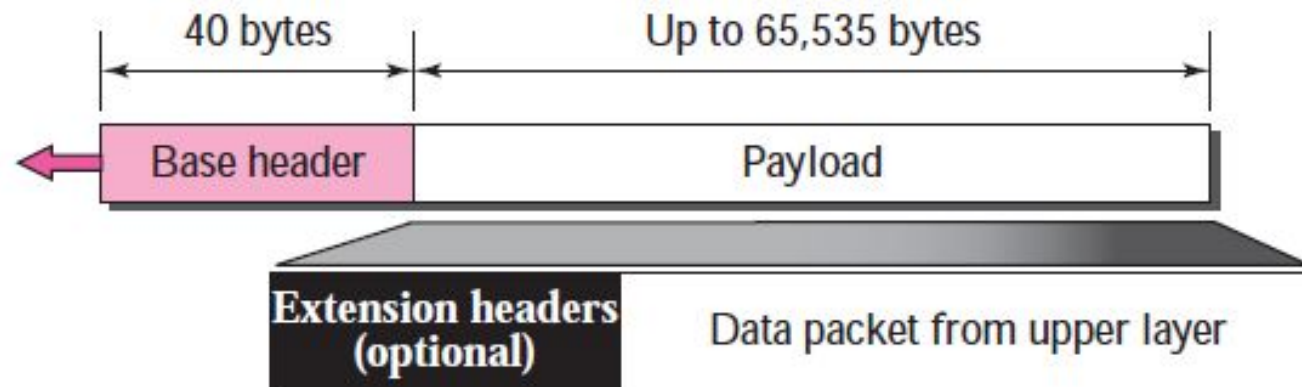
IPv6 addressing

- IPv6 uses hierarchical addressing
- It is classless and uses CIDR notation
- Example

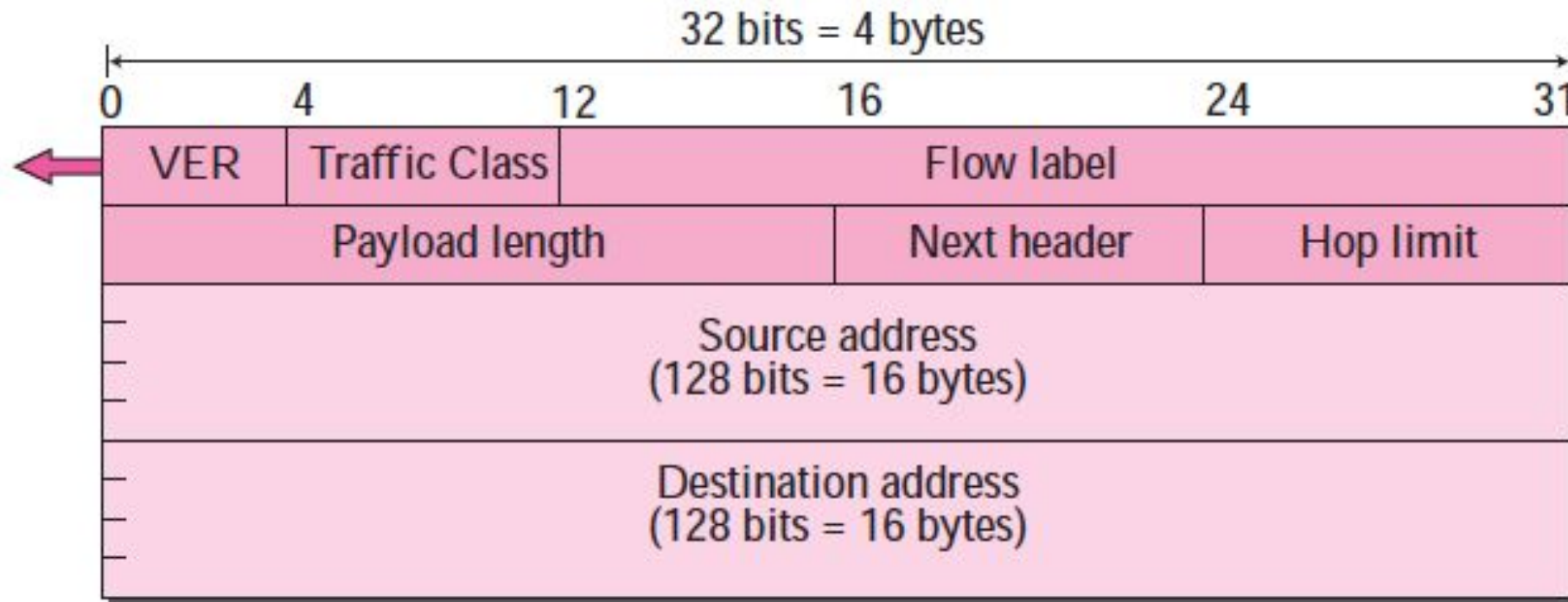
FDEC :: BBFF : 0 : FFFF/60

IPv6 packet format

- Each packet is composed of a mandatory base header followed by the payload.
- The payload consists of two parts: optional extension headers and data from an upper layer.
- The base header occupies 40 bytes, whereas the extension headers and data from the upper layer contain up to 65,535 bytes of information.



Base header



Base header field

- Version. This 4-bit field defines the version number of the IP. For IPv6, the value is 6.
- Traffic Class. This 8-bit field is used to distinguish different payloads with different delivery requirements. It replaces the service class field in IPv4.
- Flow label. The flow label is a 20-bit field that is designed to provide special handling for a particular flow of data.
- Payload length. The 2-byte payload length field defines the length of the IP datagram excluding the base header.
- Next header. The next header is an 8-bit field defining the header that follows the base header in the datagram. The next header is either one of the optional extension headers used by IP or the header of an encapsulated packet such as UDP or TCP. Each extension header also contains this field. Note that this field in version 4 is called the protocol.
- Hop limit. This 8-bit hop limit field serves the same purpose as the TTL field in IPv4.
- Source address. The source address field is a 16-byte (128-bit) Internet address that identifies the original source of the datagram.
- Destination address. The destination address field is a 16-byte (128-bit) Internet address that usually identifies the final destination of the datagram. However, if source routing is used, this field contains the address of the next router.

Software Defined Network (SDN)

Traditional Network Management

- Use SSH to manage one device at a time
- Command-line interface
- Very slow, error-prone (accidental errors, different sysadmins may configure differently)
- Later some GUI tools were available, but they were slow and inefficient

Network Automation

- Device configuration
- Initial device provisioning
- Software version control
- Collecting statistics from devices
- Compliance verification
- Reports
- Troubleshooting

Network Automation Benefits

- Enhances automation, reduces manual efforts
- Human errors and typos are avoided
- Scalable- not one device at a time
- Version/configuration control

Automation Options

- Python scripts
- NETCONF
- RESTCONF
- Ansible
- Puppet
- SDN
- Cisco DNA Center
- Note: Not all methods are supported by all the devices

SDN on Routers

- SDN separates data and control planes
- Three planes will be created
 - Data plane – Traffic of data which is forwarded through the device.
 - Control plane – Makes decisions about how to forward traffic. Control plane packets such as routing protocol or spanning tree updates originate or terminate on the router itself. Runs in a separate centralized SDN controller.
 - Management plane – The device is configured and managed here. E.g. by using Telnet or SSH or via a GUI using HTTPS, etc.

OpenDaylight

- https://www.youtube.com/watch?v=rAm48gVv8_A
- https://youtu.be/YN_nK3w51Ag

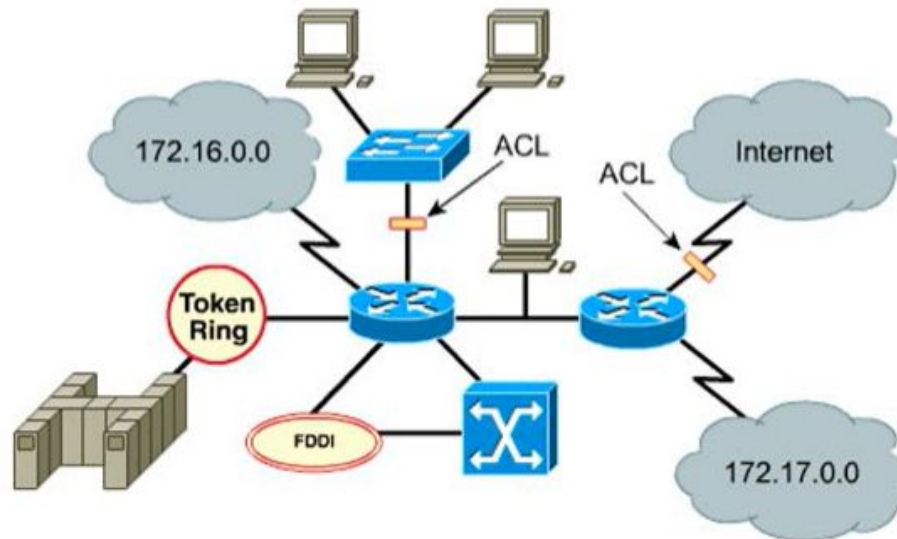
Access Control Lists (ACL)

Access Control Lists

- **Access Control List (ACL)** is a list of instruction that we can apply to a router's interface
- This list tells the router which packets to accept and which ones to deny
- Parameters for acceptance and denial
 - Source address
 - Destination address
 - Port numbers

ACL Usage

- An ACL filters network traffic by controlling whether routed packets are forwarded or blocked at the router's interfaces
- A router examines the conditions specified in the ACL to decide whether to forward or block packets



ACL Creation and Usage

- For one protocol, there should be one ACL
- Example: If a router interface is configured for IP and AppleTalk, we will need two ACLs, one per protocol
- Reasons for creating an ACL
 - Limit network traffic and improve its performance
 - Provide traffic flow control
 - Provide basic security to a network
 - Decide what traffic to allow/block at the router interface level
 - Packet filtering

ACL – Order of statements

- It is very important to place the statements inside an ACL in a correct order
- When deciding whether to route or block a packet, the router software tests the packet against the condition in each statement in the order in which it is created, sequentially
- When a match is found, no more statements are checked
- If we need to add some statements to an existing ACL, we must delete the ACL and recreate it with the new statements

Standard Numbered ACL

- The standard access list has a number range from 1-99 and 1300-1999.
- It specifies permit/deny traffic from only a source IP address.

Extended ACL

- Although there are times when we only need to filter traffic based on the source address, more often than not we will need to match traffic with a higher level of detail.
- An option for more precise traffic-filtering control would be an **extended ACL**.
- Here, both the source and destination address are checked. In addition, you also have the ability to specify the protocol and optional TCP or UDP port number to filter more precisely.
- Extended ACL range is 100-199 and 2000-2699.

Application Port Numbers and ACL Keywords

Application	Port	ACL
FTP	TCP 21	ftp
SSH	TCP 22	ssh
Telnet	TCP 23	telnet
DNS	TCP UDP 53	domain
TFTP	UDP 69	tftp
HTTP	TCP 80	www
NTP	UDP 123	ntp
SNMP	UDP 161	snmp
HTTPS	TCP 443	https

ACL Example

```
R2(config)#
access-list 100 deny tcp 10.10.30.0 0.0.0.255 gt 49151 10.10.20.1 0.0.0.0 eq 23
```

No.	Action	Protocol	IP	Source Wildcard	Qual.	Port	IP	Destination Wildcard	Qual.	Port
100	deny	tcp	10.10.30.0	0.0.0.255	gt	49151	10.10.20.1	0.0.0.0	eq	23

Examples of ACL Commands

10.10.10.10 = A specific host
0.0.0.0 = Means this is a host, not a network

10.10.10.0 = Network
0.0.0.255 = All hosts on that network

- `access-list 1 deny 10.10.10.10 0.0.0.0`
- `access-list 1 permit 10.10.10.0 0.0.0.255`
 - Default wildcard mask for a standard ACL is 0.0.0.0, which means an individual host. So the first line will deny access to the host 10.10.10.10. The second line will allow access to all the other hosts.
- `access-list 100 permit tcp host 10.1.1.1 host 10.1.2.1 eq 80`
 - Permits http traffic from host 10.1.1.1 to host 10.1.2.1
- `access-list 100 deny ip host 192.168.1.1 host 192.168.3.1`
- `access-list 100 permit ip any any`
 - The first statement denies all application traffic from host-1 (192.168.1.1) to web server (host 192.168.3.1). The **ip** keyword refers to Layer 3 and affects all protocols and applications at layer 3 and higher. The last statement is required to permit all other traffic not matching.

Examples of Extended ACL Commands

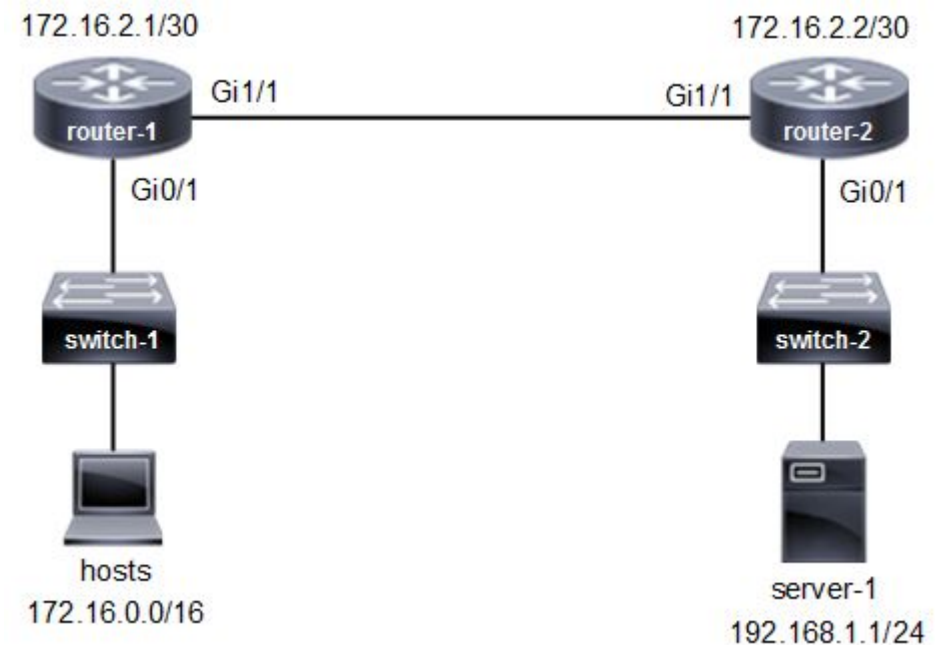
- `access-list 100 permit tcp 192.168.1.0 0.0.0.255 any eq telnet`
- `access-list 100 permit ip any any`
 - The first statement permits Telnet traffic from all hosts assigned to subnet 192.168.1.0/24 subnet. The **tcp** keyword is Layer 4 and affects all protocols and applications at Layer 4 and higher. The **permit tcp** configuration allows the specified TCP application (Telnet). The **any** keyword allows Telnet sessions to any destination host. The last statement is mandatory and required to permit all other traffic.

Examples of Extended ACL Commands

- `access-list 100 permit ip 172.16.1.0 0.0.0.255 host 192.168.3.1`
- `access-list 100 deny ip 172.16.2.0 0.0.0.255 any`
- `access-list 100 permit ip any any`
 - The first ACL permits only hosts assigned to subnet 172.16.1.0/24 access to all applications on a server (192.168.3.1)
 - The second statement denies hosts assigned to subnet 172.16.2.0/24 access to any server. That would include any additional hosts added to that subnet and any new servers added.
 - The last ACL statement is required to permit all other traffic not matching previous filtering statements.

Mapping of port number to application

- access-list 100 deny tcp any host 192.168.1.1 eq 21
- access-list 100 permit ip any any



- Result: Deny all FTP traffic from any subnet that is destined for server-1.

Example of Extended ACL Commands

- Write a command to allow telnet traffic from host 10.1.1.1 to 10.1.2.1
- Answer: `access-list 100 permit tcp host 10.1.1.1 host 10.1.2.1 eq 23`
- What access list permits all TCP-based application traffic from clients except HTTP, SSH and Telnet?
- Answer: `access-list 100 permit tcp any any neq 22,23,80`
- What access list denies all TCP-based application traffic from clients with ports higher than 1023?
- Answer: `access-list 100 deny tcp any any gt 1023`